Ep# 15 | WTF is Climate Change? Nikhil ft. Sunita, Bhumi, Navroz and Mirik

[Speaker 2] (0:00 - 4:15)

Everybody has heard of the word climate change, but when I asked him to describe it to me or tell me what it is, not many are able to. Ready, Meghna? Cut.

Hi guys, welcome everyone. I've not done this for a while. We took a break of sorts.

The very first thing we do when we begin is introduce each one of you. Who would like to go first? Sure.

So tell us a bit about you, where you grew up, how did you get into the domain that you are in right now? And just highlights and low points in your life so far. Okay.

Okay. Well, for those of us who've been around for a little bit longer, it may take a little longer, but I'll try and keep it brief. You know, I followed the standard South Asian path, especially as a South Asian male.

I started out as an engineer, but I really didn't enjoy it. I started enjoying my political science, politics, history, all this stuff much more. And so I started getting interested in development.

And I had the opportunity when I was studying, it was the late 1980s. There was the sort of emergence of discussions around the environment. And I took myself off to the Narmada Valley.

And I introduced and interviewed people who were associated with the Narmada Bachao Andolan. And many people from my generation actually got into environmentalism through that. It was a very high profile movement.

Do you actually, you know, do you have to break eggs to make omelettes? Is that inevitable consequence of development? Or can you do it in a way where you actually don't harm poor people, where you don't dam rivers, where you don't despoil the environment?

And I thought this was fascinating. So this was my first kind of entry point. And I wrote a thesis about this and so on and so forth.

And then when I graduated from college, and this was in the US, I had this, I was just in the right place at the right moment. I'd also done some work on climate change. It was 1990.

And there were preparations to start negotiating a climate change convention. So I got in touch with an organization who was involved in the periphery of the Narmada movement. And I said, look, I'm looking for a job.

And they said, well, as it happens, we need somebody to help build a global network on climate

change called the Climate Action Network. And are you interested? We don't have very much money.

So I was fresh out of college. And I was like, well, yeah, sure. And so I called up Sunita and Anil Agarwal at CSC.

I called up people in Latin America. It was the old days. It was fax machines, photocopiers.

I didn't really know what to do in an office. You know, I had a filing cabinet and I had two files, faxes in and faxes out. I didn't really know what I was doing.

You know, I used to read the newspaper under the table because I had never been in an office before. But over the next couple of years, we built up this network and civil society organizations from Africa, Asia, Latin America all showed up. And that organization exists today.

It's called the Global Climate Action Network. It's now hundreds of people. And the key part was that it brought representatives from the developing world to this global process, because until then, it had only been Americans and Europeans.

Right. So development issues emerge as part of the conversation. And I'll just give you one anecdote from that period.

Very early on, we sat around the table with the Americans, Europeans, people from Malaysia, India, everywhere else. And the Americans who'd been at this for a while said, OK, we need to come up with a common lobbying position. So what we should say is that we, the richer countries, have caused most of this problem.

So we should reduce, you know, say, 50 percent by the year 2000. This was in 1990. Very ambitious.

[Speaker 3] (4:16 - 4:16)

Right.

[Speaker 2] (4:16 - 9:16)

And the developing world will do the same thing five years later. And I still remember a colleague from Malaysia said, wait a minute, but our per person emissions are so much smaller that that would lock us into a much lower level of emissions in perpetuity. And the Americans sort of scratched their heads and they looked at each other and they realized there wasn't a common ground.

And the funny thing is, in many ways, we're still having the same conversation today, 30 years later. Right. So this was my sort of trial by fire.

I did this for two years and then I decided this is sort of crazy. I know nothing about development. I know nothing about the world.

I'm sitting at these U.N. negotiations just because I was in the right place at the right time. I need to step back. And so I went off to do my PhD.

And for that, I started looking at water use in Gujarat, in two villages. So I spent a year in these two villages. So the other extreme, from the global straight to the sort of village level.

And I studied this as an academic and thinking about what one does about it. And so those two experiences were really kind of shaping and framing. So I'm not going to give you the blow by blow, but sort of fast forward after working for a little while in the U.S. I came back to India in early in the early 2000s, bounced around a little bit, universities, this, that and the other. And then I landed at a wonderful institution, the Center for Policy Research. And there I built a team of people working on energy, climate change, environment. I was able to collaborate with Sunita and other colleagues and other organizations and have been there for a decade until just last month, when I left along with our group to set up a new institute called the Sustainable Futures Collaborative.

And we are just getting going this year. So that's the arc. Tell us a bit about IPCC, which is, I think for many people, the Bible of this industry.

You're a co-author. So as part of my work at CPR, I was, so, you know, I both publish and I do policy work, right? So as part of the policy work, we get invited to sit on government committees and so on and so forth.

But a core belief of our group is that we have to be, our work and our policy positions need to be based on rigorous work. And therefore, publication is an important part of that. So in that context, I got engaged with the IPCC, the Intergovernmental Panel on Climate Change.

So it's a really strange beast, right? So it's three working groups, science, impacts, and what are called response strategies. How do you respond to it?

And I was in the third of these, and I did it twice. Each cycle takes seven years. So the way it works is you don't come up with new information, but your group is meant to be the authoritative compilation of existing information.

So I was tasked this last time around with the unbelievably impossible task of compiling all the information for Chapter 13, one of 17 chapters. And my chapter was called National and Subnational Policies and Institutions. So all the policies and all the institutions needed to address climate change.

And we had to pull that together over the COVID period through Zoom calls, you know, with

authors all over the world. And it's a very, very complex process because governments take this really seriously, right? So governments at the end, this isn't just a research report.

Thank you. This isn't just a research report because by the end of it, governments approve this report on a line by line basis. They approve the summary of it.

So they literally put up on the screen one sentence at a time. Governments will comment on it, suggest modifications. The authors have to then go back and say, this is consistent with the science.

This is not consistent with the science. And they're jockeying for position. So it's an incredibly complicated and incredibly painstaking process.

So the approval happens over a week where you're basically locked into a room working 12, 14, 16 hour days with all the government delegates. And very slowly, very incrementally, you come up with formulations about is climate change real? What do we do about it?

What are the likely impacts? And they put probability metrics by it. This is 50 percent likely.

This is 90 percent likely. This is highly likely and so on. So it's a very unwieldy process.

And we'll talk about the negotiations later, I'm sure. They're both incredibly unwieldy processes. And I've had thoughts more than once that you're dealing with this urgent, dramatic situation.

And sometimes I feel you couldn't design a more cumbersome process globally to deal with this.

[Speaker 1] (9:16 - 9:17)

And you review it every seven years?

[Speaker 2] (9:18 - 11:30)

Every seven years, there's another cycle. A new set of authors are brought on and you start the whole thing over again to work with them. Sneeda, would you like to go next?

No, let me go first. Sure. So I think just a brief about me.

I'm born and brought up in Mumbai. So I studied here also. So I never sort of went out of the city.

I think early days, so when I was like four, four and a half years old, I had a brain tumor actually. And that got, so that was a challenging time. I don't remember it about it too much, but at least for the family and everyone around, it was a challenging time.

But luckily, no major issues after that. But that sort of shaped my thinking later on. So I come from an engineering family.

My father is an engineer, my uncle is an engineer, elder brother is an engineer. So that whole engineering mindset that you want to go into engineering. So then I did preparation for IIT.

The first attempt was not a very good attempt. So I didn't get a good rank, took a drop, did the whole thing again. And then again, had a target to go to IIT Bombay.

I had an option between IIT Bombay and UCLA in the second attempt. And then I said, look, I had heard so many stories about IIT Bombay from my brother. The mood indigos, the hostel life and all that.

So I said, look, this is the place. And then during IIT, at the end of IIT, it was the startup phase. If you remember, like all the, it was 2013, 2012.

How old are you now, Vivek? 33. What year are you talking about, IIT?

So yeah, so I passed out in 2013. So 2008-13. So this was a time when the housings of the world and tiny houses of the world.

So all sort of my batchmates were either founders or startup founders. And there was a whole, at that time, the hope of Powai Valley being created. So I also had my own journey there with my batchmate Ayush Jain.

We started this company called Humming Whale, which had nothing to do with sustainability at all at that time. It was just a pure play. Let's create cool things and then license them or collaborate with big corporates to make them a successful.

And your aim is just to create cool things. So over between 2013 and 16, we would have made like 15 odd products.

[Speaker 1] (11:31 - 11:33)

What are these cool things? I'm very curious.

[Speaker 2] (11:33 - 14:26)

So one, I think the most famous one was we made a cricket bat, which was more aerodynamic in nature and the edges wouldn't travel to the slips. And that was in reaction to one of the India's test matches in England, the test series where we lost 4-0 or something. And we were like, okay, can we do something about it?

So in IIT we had all these people who could do the simulations on softwares and a lot of iterations, 10 iterations. We got it approved from the Mary Bloom Cricket Club. We sent it there.

They said that it's legally allowed. Then we met everyone like at that time Sachin or Rahul Dravid or Kohli, Pujara. So it was an interesting time.

For whatever reasons, it didn't scale up as we hoped for. We sold a couple of hundred bats and some more interesting things like that. Then we realized, look, this is a passion project.

It's not really going to turn into a business. I think business will require the depth, the focus. And at that time, when you look at the total addressable market of a cricket bat in India, it was like a thousand crores.

So not much of a dent that you could do in it. In 2016, we took the call that let's shut it down. And from 2016, I joined Aarti Industries, which is sort of my company that my father and uncle founded in 1984.

So it's going to be 40 years now. And I think for me in the climate journey, around 2019, 2020, just before COVID, so I do read a lot. And I went into the rabbit hole and Google and the YouTube algorithms really trapped me into that climate phase and did a lot of courses.

So a lot of self-reading. And earlier, it was more about just understand the whole problem, not really do anything about it. But then I actually had gone to COP26.

And before that, we did some interaction with a few organizations globally. And in COP26, for me, it was like a sense of urgency hit me, like this is the decade, the decisive decade and all those things. And I said, look, we really, I came back, talked to the family that we really need to start within our sphere of influence, whatever that is, work with organizations and make our sort of point or make our impact in the overall climate story.

So yeah, I think now we are closely working with over 60 organizations globally in various sectors, including Navrose, but others as well. So me and Merik have not known each other for too long. Last year, we were in Goa at an event.

And Nandan put us on a table, like four of us together. One of them was a geothermal guy. One was a geoengineering guy.

One was the UCLA professor for climate. And Merik and I. And we didn't expect it, but we all got along so much in this event.

And we had dinner and we hung out that night. And then we just kept in touch. And you made a WhatsApp group after dinner saying climate podcast.

[Speaker 3] (14:27 - 14:28)

That's so cool. I'm quick.

[Speaker 2] (14:29 - 16:30)

Yeah. Because when you're getting so much information from different people in the same

industry, it has different perspectives. I think that's very interesting.

What I also found interesting about Merik is because of Aarti Industries, they actually are, it sounds bad when I put it that way, but you're running large equipment, which is producing a lot of emissions. They can be the first-hand people in making significant change. If industries were to incorporate new technology that is coming about.

I think one of the thoughts from putting the corporate sort of thought process is we are into the chemical industry. Chemical industry is a high energy consumption industry. And we have been doing our sustainability reports now.

So our emissions would be, scope one would be like 650,000 tons, 680. Scope two will be 140. And just to give you a sense, scope one is the emissions within your own premises and scope two is the bought out electricity.

And if we are able to use some of, engage with early stage companies and try to scale up their solutions, it obviously benefits us. But it's a challenging problem in running an industry where coal is an energy source. How do you get away from it?

And we are also looking at multiple solutions to do that. But yeah. Also another request to everybody.

The target audience for this, in my opinion, is the youth of our country. People in the age group of 16 to 35, 40. So as far as we can avoid it and not say anything complicated or use jargon or things that a 16-year-old boy or a girl can easily understand.

I think we should stick to that. I think that's the opportunity largely. These are the people who will build for tomorrow.

And I think we need to focus on that. Who would like to go next?

[Speaker 1] (16:30 - 18:50)

So I'm Bhoomi. I'm an actor. Don't ask me my age.

But I live in Bombay, born and brought up. I am also very passionate about everything, all the conversations, all the action around climate change. I kind of want to dig into where this started from.

I remember we'd learned about climate change in school. And I was like, OK, you know, our environment's changing. Our weather is changing.

No big deal. But obviously, like, you know, we were told that our fresh water is going to dry up. You know, there's going to be drought.

Like, it's going to be quite apocalyptic. And then I watched a film called The Day After Tomorrow. And I thought, this is real.

This is going to happen. And I was very young. I was probably like in my preteens.

And I was like, this is going to happen. And I'm going to drown. My family is going to drown.

My city is going to drown. And, you know, how am I going to save them? So I think that that movie impacted me so deeply that I kind of started reading up.

I started questioning my elders, people around me. And nobody seemed to care. You know, nobody seemed to care.

They were like, no, it's all OK. It's just a conceptual fact. You know, it's not going to affect you.

It's going to happen like in the future. Your children are going to outlive it. You know, we are not really going to see it.

And I was like, OK. But then as I kind of went on in my journey, I realized that it's happening now. Right.

And it bothered me. I actually had like many, many anxious nights. And because it felt very lonely at the start of it.

Because if I ever went and I spoke to anybody that, you know, I feel this way, like it bothers me. You know, what's going to happen? I was made fun of.

Nobody really kind of... They couldn't comprehend what my thought process was or where my fear came in. And that's when I kind of went in looking for like-minded people.

There weren't many. I wouldn't lie. This is like maybe...

And then I became an actor. And I realized that I have...

[Speaker 2] (18:50 - 18:52)

You wanted to work on climate before acting?

[Speaker 1] (18:52 - 19:02)

Yes, way before acting. You know, I was that child that if there was a national disaster, I would be on the roads collecting chanda. So I think I've always been like an empathetic person.

[Speaker 2] (19:03 - 19:45)

I'll pay Bhoomi a compliment. Not because she's sitting in front of me. But when COVID was

going on, we didn't know each other and somebody had connected us.

And I was like Bollywood star, you know, like all of, you know, what would we talk about really? But every time we spoke, Bhoomi would always call me and be like, this person in this remote part of Bangalore is looking for an ambulance or they're looking for a hospital bed. And she would join all the calls around this very diligently for a very, very long time.

And I was so impressed. I was like, she's not in Bangalore. She works in Bollywood.

She's an actress. Why is she calling me to find an ambulance for somebody in remote Bangalore? Which was very impressive.

[Speaker 1] (19:46 - 21:01)

Thank you. That was quite an experience. I mean, you did your fair share, which was so, it was...

Yeah, I think like everybody did their bit. So just going back to what I was saying, then my acting career started and I realized and I've always kind of stood for films that make an impact. And I realized that, okay, now I have the power to reach out to a larger audience.

So, you know, might as well use it to advocate for any kind of positive impact that I could. And I started Climate Warrior in 2019. Through Climate Warrior, we did a lot of on-ground work, which were like beach cleanups, plastic collection drives, talks, panels, wherever my voice could reach and whatever change we could get.

Our target audience was pretty much the youth of India. Because as you rightly said, they are the change makers. They are the ones that are going to find the solutions.

It's been quite a satisfying journey since I started Climate Warrior because I met a lot of like-minded people. So suddenly it didn't feel as lonely. And I was like, okay, there are people that understand why I get anxious or, you know, there are people that actually want to bring about a change.

And yeah, that's why I'm here.

[Speaker 2] (21:01 - 21:01)

Sweet.

[Speaker 1] (21:01 - 23:30)

Yes. So in fact, a lot of what you said, Bhumi, resonates with me. But strangely enough, even though I'm so much older than you, so when I started out, and that's a long time ago, 1980, nobody knew the word environment.

Right. Okay. And I had this wild idea that I was going to work in the field of environment.

People ask me, why? And I still don't know. But I think it goes back to my mother who was sort of Delhi's Navrosni, who was Delhi's most avid gardener.

And so, you know, I grew up with green all around me. And at that time, nobody really knew what the word environment was. And at school, it was very strange because there are three of us who met, you know, life is about coincidences.

And we met at this strange conference in Delhi, which was organized by now, what I say, the grown-ups. I mean, at that time, we said the grown-ups talking, talking, talking and not doing anything. And we all ended up there.

And we set up, we decided to form a group just like you. But this is 1980. Wow.

It was called Cultrich. It was a student group. We were all in school.

I was in modern school in Delhi. And there were others in school. And we were all, we used to just meet and take up causes.

And we've took up causes all the way from protecting trees that I still pass that road every day. And I think about it that, so I don't know if you've been to Delhi, but there is this very, there's this road that leaves from Safdarjung tomb down to IIC. You know, that main, what's called the boulevard.

Okay. And at that time, they were going to chop down all the trees. And it was our fight, which actually, and this is 1980, 79, 80.

And at that time, so they decided to keep a few, the central verge. So there is a central verge of trees. And there is then the road on both sides.

Small victory, but it feels good. And we set up a group. And like Navroz went to the next big campaign that was hitting India.

I went to the one before. I went to Chipko.

[Speaker 2] (23:30 - 23:35)

I should just, sorry to interrupt you there, but Kalpavriksh is who I went to Narmada with.

[Speaker 1] (23:35 - 24:12)

Yeah, I know, I know, I know. That was the soul of the movement at that time. So I went to Chipko and that was, you know, I'm elite to school.

What do I know about trees? I mean, for me, environment is about protecting trees. And you go to this place where there's this very, you must go.

I mean, though the village is now finished, the women are gone. It's a very sad story, but still, I mean, something I never forget, Humi, that one, I go to this village where women have done what is now known as the Chipko movement.

[Speaker 3] (24:13 - 24:18)

And there was a movement, you know, to protect. So the Chipko to the trees.

[Speaker 1] (24:18 - 25:27)

Now, as somebody coming from Delhi, you know, the whole idea was, oh, they're going to, they're protecting their trees. What lovely people. And you go, you know, my first, I was brought down to earth and, you know, really slapped in my face when I, you know, when I said this to the women and they said, oh, we're not protecting trees.

We're protecting our toilet ground. This is where we go in the morning. And if those trees get cut down, where should we, where will we go?

We are protecting our right to cut the trees. I can never forget those, those words, because for me, that changed my whole idea of environment. It became from this sort of woolly headed, you know, let's protect trees to, this is about development.

This is about the right to development. This is about poorest women in India, way up in the Himalayas, talking about why they need to protect the trees, because it's about their survival. And that story, and I'm sure it was the same in Armada.

That story is, those stories don't leave us. Those images don't leave us.

[Speaker 2] (25:27 - 25:41)

Sunita, if I may ask, all of you are talking about childhood and getting onto this path that you are on today that early. Do you think psychologically a certain kind of person wants to do a certain kind of thing that early?

[Speaker 1] (25:43 - 25:58)

It's difficult. I wouldn't know. I find my life is about bumbling on.

I just got to it. I enjoyed it. I stayed with it.

I can't, when I look back, I have no regrets that why did I go to Chipko?

[Speaker 2] (25:58 - 26:02)

What was the trigger? What took you there in the first place as a kid?

[Speaker 1] (26:03 - 26:08)

So that's what I'm saying. A lot of people have asked me that. Honestly, Nikhil, I don't know why I got into environment.

[Speaker 2] (26:09 - 26:12)

And what was the high that kept you continuing it?

[Speaker 1] (26:12 - 26:25)

I think just the challenges, even now. I mean, both the challenges, the, I mean, as you say, on one hand, we are frightened to death. On the other hand, there is this enormous sense of potential.

[Speaker 3] (26:26 - 26:26)

What can happen?

[Speaker 1] (26:27 - 26:47)

What can be done? And actually, a lot of my life has been pushing. And Navroz knows this because I often am the crazy woman because I push a lot harder and a bit louder.

And but I think that's what it's all about.

[Speaker 2] (26:47 - 26:48)

Can I ask you another question?

[Speaker 1] (26:48 - 26:49)

Yes, of course.

[Speaker 2] (26:49 - 27:07)

They say bravery today is not doing what the public wants you to do, but doing what the public doesn't want you to do. Are we at a point on the planet where what the public wants to do is save the environment or save climate or change?

[Speaker 1] (27:07 - 27:42)

You've hit the nail on the head, Nikhil. I think today it's important for us, and I hope we'll come to

that conversation. Because I think today, when I think back on it, I think we were braver 30 years ago.

We had the ability to think differently. We had more courage. I think now there is a lot more talk and less courage, less imagination.

And I think everybody talks about climate change. Because it's cool now? It's to some extent cool.

To some extent it's real. There is no doubt about it.

[Speaker 2] (27:42 - 27:48)

Is it one of those things that the people who are talking most about climate also understand it less?

[Speaker 1] (27:49 - 29:15)

No, I'm not sure I would say that. I just think it's a... I think today there's just a lot more talk.

It's a cacophony. Everybody's talking. But if I, and I'm sure we will discuss it, where is the action?

I mean, when we pushed for clean air in Delhi, for instance, I was 15 years, 20 years actually, on a court-mandated committee. I had an enormous luxury. I reported directly to the judge.

And I was on a committee, which Navroz joined also for a while. And I basically, if he gave a report, it was listened to. And we pushed.

We got CNG in Delhi. And that was very brave because there was no CNG across compressed natural gas. So essentially changing the fuel from diesel to gas.

We got diesel. Diesel was our bugbear. We couldn't get it banned, but we got the price differential change.

So, you know, a lot of pushing and inconvenient things were done. And it is inconvenient. The point today is we want climate change sorted out.

We want the world's existential issue to be solved. But we want to do that at no cost. No cost to us, no cost to anyone.

And that's not going to happen.

[Speaker 2] (29:15 - 29:51)

Could I just come in on this question of bravery? You know, I think I see it probably, bravery is definitely part of the equation, but I see it slightly differently. And perhaps this also, you know, parallel tracks that Sunita and I have forged in our career.

I ultimately feel that a lot of the challenge is re-imagining the problem and communicating it in a way that you bring the public with you. Because if you want lasting solutions, you have to have the public on board. You have to have support for what you do.

And I think that's partly what's...

[Speaker 1] (29:51 - 30:17)

But Navrooj, the point is you can't stop there. You have to re-imagine. Yeah, it takes...

You have to bring people on board, but then you have to stand behind the solution and see it work. I agree. Today, we tend to be...

I mean, everything is about optics today. I'm sorry. I'm sounding a little...

I mean, I just think we need to move a little bit more behind. I mean, you want e-vehicles, get it done.

[Speaker 2] (30:17 - 31:51)

I think what ends up happening, though, is it's also therefore really important to work in partnerships. You have somebody who kind of cracks open the conversation and it requires that sort of hard blow. You have other people who kind of reframe and reshape the conversation.

You have other people who are much more solution and implementation-oriented. So I think there's a role for all of those things. Everyone.

And I think the way I've seen my own career is in trying to build that consensus and build support for a way of re-envisioning it. Just to come back to climate change, right? During my main effort during the decade of the 2010s was to say in India, yes, there are these climate changes, this terrible issue in terms of equity, right?

It's absolutely the case. But it's also the case that we in India have to now start thinking about what climate change means for us ourselves in terms of adaptation and in terms of mitigation and using this language of, it's a bit jargony, but co-benefits. Things that are good for development and that will also bring climate gains.

Is there such a thing though? There is public transport, right? We're not going to build cities in India without public transport.

We're already at density. Decentralize electricity because you're solar. All sorts of things.

You can decentralize the electricity or energy source, which until now, if you had a concentrated energy source, you couldn't. So I think some of these technologies can have co-benefits in that way as well. The tricky thing is that you don't want to use that.

You don't want that to become an argument for rich countries to escape their responsibility. So that's what makes it really, really hard.

[Speaker 1] (31:52 - 32:20)

But I have a very basic take on this, right? Like when we speak of our public, okay, if I speak specifically of our country, a large portion of them don't even know that there is a thing called climate change and that they are dealing with. That is a certain section of the society.

I'm sorry, Bhumi. I totally disagree. This is where you have to separate out the middle class from the rest.

[Speaker 3] (32:20 - 32:20)

Absolutely.

[Speaker 1] (32:21 - 32:41)

The rest of India knows climate change. And which is what? Which is the rest of India that you are specifically speaking for?

The farmers, the fishermen, the nomads, the people who work on land. The people who deal with the land, they understand. They understand.

They may not have the fancy word that you and I have and call it climate change. That's actually what I was getting to.

[Speaker 3] (32:42 - 32:44)

That's right. Then I interrupted them. Please go ahead.

[Speaker 1] (32:44 - 33:27)

They are the ones, there's a certain section of the society that is actually dealing with the adverse effects. They don't know what's happening, but unfortunately they are not empowered to do anything about it. That's when we kind of reach a higher realm, right?

How do we make sure that the people that are actually being affected by it in a larger way find a solution is what I was actually getting to. And Bhoomi, that's the global politics actually what Navroze is talking about. Because, I mean, you know, it's the powerlessness of the people of the South to be able to really talk about the fact that...

[Speaker 2] (33:27 - 33:28)

Global South, not the Indian South.

[Speaker 1] (33:29 - 35:20)

Not the Indian South. Thank you, thank you, thank you. This is one of those jargons that we get so used to.

So thank you for pointing that out. The fact is that if today you have this very unequal balance where very few people have created the problem, continue to emit, but the large numbers of people who should have a voice don't have a voice. And that's frankly the issue in India as well as the globe.

And I think that's where in some senses we need to look for solutions. We need to look for solutions in India, which are different. We need to stand behind the solutions and we need to scale them up.

And my biggest bugbear is that we are still thinking very small in terms of the scaling of the solution than the speed of the solutions that we need. I mean, when we moved to compressed natural gas in Delhi, CNG, we brought in 100,000 new vehicles in gas in a matter of a year. Wow.

In a year. We built the infrastructure, we got all the gas supplied. Tata and Leyland produced the buses.

They'd never done CNG buses before. Set up a factory in Alwar, set up a factory to produce those buses. Those buses rolled in.

You could see clean air in Delhi for a period, for a small period, because the scaling of the solution was big enough to have an impact. You know, my issue today is that we are talking about e-vehicles, for instance, and we're all passionate about it. Very much part of the solution, has to be a part of the solution, particularly e-buses rather than e-cars.

But we are moving on it at a scale which is just, we need to do this.

[Speaker 2] (35:20 - 35:27)

Can I digress a little bit and ask you a question? For all the young people out there, I'm not young, but I'll add myself into that category.

[Speaker 1] (35:28 - 35:29)

No, younger than me.

[Speaker 2] (35:29 - 35:41)

People looking to invest in electric vehicles or people looking for jobs in the EV industry, how hopeful are you of that industry from a 10-year lens?

[Speaker 1] (35:42 - 38:18)

So, I'm very hopeful that it's going to happen because it has to happen. Let me put it that way. The imperative is clear.

Why do we need EV? Let's also understand that because the rest of the world is very romantic about EV. I mean, EV in the Western world is largely an electrification issue, that the more you electrify and clean up your electricity source, the better it is, rather than having those decentralized, many sources using oil.

For us, the EV story is slightly different. For us, we need EV to clean up our air pollution. I mean, if we get EV into our cities, we will dramatically reduce, improve air quality and you know, Mumbai, Delhi, now almost every city of India is choking, absolutely.

So, EV for us is an imperative. But there are two issues, which I think we need to address and address them very clearly. And one is, of course, the war that is happening between the world, the US and China, which is really about the control of the supply chain, which is new minerals, technology, battery technology, lithium, cobalt, graphite, all in China.

When climate change began, Nikhil, and it's just a little bit of a digression here, it's important to understand the role of trade, because for most of us, when we looked at climate change, we never saw that there was a parallel thing happening in our world, which was WTO and free trade was beginning to happen. And as soon as China joined WTO in 2002, their emissions skyrocketed. And that's the reason why.

Can you elaborate each of these things? WTO is? So World Trade Organization, it's basically an agreement.

So you have two agreements. You have a global agreement to save the planet under climate change. You have a global agreement to do free trade.

Agreement between? Between global. So this is between all countries, all nations of the world signed and China was not part of it.

China became part of it in 2002. And it's a clear relationship between when China joined and how its emission curve went up. Now, the problem is not that China's emission curve went up, but the fact the rest of the world thought they had reduced their emissions, but they hadn't.

They had exported their emissions to China because they were getting all their goods from China.

[Speaker 2] (38:18 - 38:47)

In fact, there's a graph which has consumption emissions rather than country producing them.

You look at the consumption levels and then you get a different picture because all the pollution in China or emissions are linked to consumption in Europe and US. Sunita, another question to you.

You were in the Times 100 most influential people list and you went on Leonardo DiCaprio's documentary and spoke back. Spoke back? No, how do I?

I didn't speak back. Critiqued him.

[Speaker 1] (38:47 - 38:54)

No, I critiqued. I told him what I felt about the climate change politics and how inconvenient it was.

[Speaker 2] (38:55 - 39:01)

So what is giving you the biggest high in your own life right now? Without filters.

[Speaker 1] (39:04 - 39:56)

I mean, I look for highs and good things that I can see happen. I'll be very clear, Nikhil. It's not easy anymore.

OK, for somebody who's seen, I mean, 40 years ago when I was in Rio in 1992, I keep calling it the age of innocence. We just felt it was all possible. You could go and scream, you know, down with Bush, you know, and it would be down with Bush.

I was much worse. I've become very, very gentile. You ask now, Rose, I mellowed.

So I think the high in my life, I mean, I went two weeks ago, no, two months ago, I went to these villages in Andhra Pradesh, looking at natural farming and I wrote about it. Amazing. I mean, to meet those women.

[Speaker 2] (39:56 - 39:58)

Why does that give you a high? Subconsciously.

[Speaker 1] (39:58 - 40:36)

Because I see the potential of scale. For me, I'll be very clear. Where I've reached today, I am tired of small solutions, pilots, nice ideas.

I really feel I get a high from seeing things scaled up. I'll give you another example. So I do a lot of work on excreta.

That's one of my major areas of work is shit. And we do these shit flow diagrams. I did a two

volume book called Excreta Matters, essentially arguing that if you don't get your excreta story right, you cannot understand, you cannot fix your water pollution.

[Speaker 2] (40:38 - 40:41)

I'd love to know more. How do you get your excreta story right?

[Speaker 1] (40:42 - 40:57)

OK, so I'll explain. Don't get me started on this. You'll regret it.

So the shit story is, if you look at Mumbai, we've done a shit flow diagram for Mumbai or Bangalore or any other.

[Speaker 4] (40:57 - 40:59)

Which city is better? Which one is better?

[Speaker 1] (40:59 - 43:22)

All of them are bad. All of them are bad. Most cities of India do not treat, do not have underground sewage to intercept sewage and to take it to a sewage treatment plant, clean it up and then take it to a river.

Most cities of India do what Mirik just talked about, the decentralized energy. This is decentralized sewage treatment. What you call on-site sewage treatment.

Now, what excites me about it, and I then started, I got really excited by this saying, here's the potential. I mean, if we could go to the cell phone, I mean, I'm old enough to tell you that when we were trying to get a landline for CSC, for my organization, I actually had to go to Jairam Ramesh, who was OST at that time in planning commission. And of course, Jairam got many, it was a payback which I had to, but he had to sign on a piece of paper for us to get a telephone line.

Today, all of us take that for granted. We've gone through the satellite. What Mirik is talking about is this amazing opportunity that we don't build the grid, we can go through our decentralized energy sources.

Just think about it. When it comes to rainwater harvesting, you capture the rain, you harvest the water, you'd use it. It's the same thing with sewage.

Now, what excites me is that we started this about five or six years ago, asking for what was impossible. And my biggest problem always is the engineering mindset, extremely sort of technocratic. You know, it cannot be done, madam.

Yeh toh hoi nahi sakta. You know, aap toh desh barbad kar dengi. Cholera ho jayega poore desh

mein.

You know, kya hoga? Yeh kaise ho sakta hai? Etcetera, etcetera.

Today, it's government policy to do decentralized sewage treatment. An entire state of Odisha has decided not to build underground sewage systems, but to take... So it's a very simple thing.

You have septic tanks, but nobody cared about those septic tanks. So now you have trucks lifting the sewage, which you had earlier. But now you have GPS on the trucks, take it to a treatment plant where it gets treated.

But the best thing about it is that you reuse and recycle that water and the sewage back on land.

[Speaker 3] (43:22 - 43:22)

Wow.

[Speaker 1] (43:23 - 45:56)

So you don't put it back in water. So you, you know, it's that... So, sorry, Nikhil, I've gone a little deviant, but to just explain, these things excite me.

Seeing it happen, seeing the idea take off. So I agree with Navroz completely. Imagination, thinking differently, and then getting it implemented.

So, you know, with the whole CNG movement that you did, when you did, I have, I want to ask you something. You spoke about rainwater harvesting, right? Like for a city like Mumbai, which gets like insane amounts of rainfall.

How would you turn that into policy? Like what, what does one do? And I genuinely, I'm asking you because that's something that I want to do.

So Mumbai is, see, Mumbai is a crazy city from its water and sewage point of view. You bring your water from very far away, okay? From, in fact, in the Western Ghats, you have two reservoirs from where the water is transported.

That water, if you've read, has a lot of tension because local people say, that's our water, we shouldn't send it to Mumbai. Mumbai then brings the water here and you have destroyed, I mean, I keep saying you're, you're unfortunately inheriting this, but I can call Navroz my generation. We're a generation which has actually lost rivers.

Your Mithi is declared a drain. Yes. Okay, it's declared a drain.

It was a river, freshwater river. In Delhi, we have something called the Sahibi. It was a river, a freshwater river, which came and joined Yamuna.

It's called the Najafkar drain today. Okay, you have Buddha Nala in Ludhiana, which was a freshwater river. It's called the Gandha Nala today.

You know, so, Mumbai really needs to get its act together. It can do a lot. Citizens need to talk about, you have high-rise buildings, so your problem is, you can't really do rainwater harvesting at the scale that we can do in Bangalore, for instance.

But you have lakes, you have ponds, you have ability to be able to use your rainwater because you've got a huge amount of rainwater. Your drain is completely in the sea. You need to find a way to be able to hold that rainwater back on land.

I don't know how far we are from Pobai, but that is one of your biggest lakes. I think. It is.

It is one of our biggest water sources.

[Speaker 2] (45:57 - 47:13)

So, one of the things I've realized talking to people is everybody has heard of the word climate change, but when I asked them to describe it to me or tell me what it is, not many are able to. So, the one thing we want to build is awareness. I'm going to go through each of you and ask you to define climate change in, say, two minutes.

I'll give you my very idiotic, non-aware description of it. Sun emits rays. Each surface on the planet reflects differently the rays back.

I think it's called albedo. I remember it using the word albino. When the heat is going back, it gets trapped in the atmosphere by virtue of more carbon dioxide, more methane, more water vapor, whatever, and the temperature rises.

But I want each of you, in your own words, describe climate change for me. Some definition will resonate with somebody. Should we start with Navroz?

Maybe let's start with Miric. I'm curious to hear. So, climate change is this system in which the rate of change in the temperature and ocean acidification and the other parameters that impact any locality is higher than the ability of the people or organisms in the locality to adapt.

That's a good definition.

[Speaker 1] (47:13 - 47:15)

That's a nice way to put it.

[Speaker 2] (47:15 - 47:50)

Navroz? Well, look, I don't want to repeat what Miric has said. I think he's explained the science

of it really well, right?

But I think the key point is this, to pick up on this point about the rate of change. The Earth is now warmer, on average, than at any point it's been in the last 100,000 years. If you look at any of the graphs, you just see this incredible spike right in the last 30, 40 years.

And you're able to correlate that with human involvement. Absolutely. More carbon, hotter temperature.

[Speaker 5] (47:50 - 47:50)

That's right.

[Speaker 2] (47:51 - 55:27)

Since you asked me about the IPCC, let me explain a couple of the quite amazing science, bits of science behind this. But what I really want to get to is actually to move away from the scientific definition of climate change. My own definition won't be around science.

But let me give you a couple of examples. So one of the ways in which we know that, and completely right, right? We've had these cycles, the Earth is warm, cool, et cetera, et cetera.

But there have been massive disruptions to life on Earth associated with each of these things. So we say it's natural. Sure, it's natural.

But the disruption is also natural and it affects us now as a species. So one of the things we've done is we look at ice core data, right? So they take literally cores of ice digging deep down into the Earth.

So the deepest is the Vostok ice core data, goes back 800,000 years, right? The ice core has kind of rings because every layer of snowfall has a distinct pattern. So they can count how old that ice is.

Then they look at the air bubbles trapped in that ice. They call it sedimentation or something. That's right.

So the air bubbles that are trapped in there allows them to say what was the percentage of carbon dioxide at that point? A lot of climate skepticism in the US, very, very little in India. And it's also the case- Is it also because less climate dialogue in India?

Well, I don't think that's actually the reason. There's been systematic efforts that has been documented now at disinformation campaigns, right? By fossil fuel companies.

That's a more likely explanation. But just going back to this Vostok stuff. So you know the carbon dioxide, you know the temperature when the snowfall fell, you can map them.

And you see this one-to-one relationship. Every spike in carbon dioxide, spike in temperature. So we know the two are really closely associated, okay?

Now, what have we done more recently? These incredibly complex climate models that have been refined over time. And you'll see this in the IPCC.

They say, all right, let's run the climate model as if the science of climate change was fake, was false. And you see the temperature record. So they, you know, and they map it to the previous temperature.

They calibrate against the previous temperature. And it basically looks like it's more or less, it doesn't change, right? Then they say, now let's apply it to what we think is the science of climate change and look at what it looks like, right?

And it edges upwards. And then they map real data on top of it. And the real data and the models are a near perfect match.

So we've become pretty good at actually representing the science. I think we're- Can I do a counter narrative because somebody has to, I'm like scared to get to the point where we all say the same thing. As long as we get to the social part and not just the science part of that.

Yeah, 100%. Okay. Okay, first thing, I don't understand the science, but I'm going to ask you questions about it.

I really want to know why does carbon dioxide, for example, stay in the atmosphere for 300 years, whereas methane is only around for 15 years, all of that. But people also do say that the kind of radiation that carbon dioxide is blocking is near peak and more carbon in the atmosphere might not significantly mean more heat. Is there any logic to that?

Look, so I'm going to let Meric weigh in as the sort of engineer in the bunch. My own take on this is the science of this is so incredibly complicated, right? For example, there were arguments put out about clouds and what effect the clouds would have.

And when you add up all of those things, so as a lay person, I don't think any of us can apply our own mind to the collective of the problem because there's so many dimensions of it, right? So there's the oceans and the atmosphere and the interface between them. 90% of the excess heat is sitting in the oceans.

There's the ocean cycles, right? There's the various gases. You said carbon dioxide, methane, different lifetimes.

And I couldn't honestly tell you exactly what the atmospheric chemistry is of why there are these different durations. This is the kind of area where you have to look at professional scientists and say, give us a good synthesis and give us evidence, right? As to why you're coming up with the

conclusions that you have.

And the IPCC, so I've been in that process, right? Every chapter, my chapter had 1,200 comments times three in three towns of review. So the best that we have is critical scientific review.

Is it perfect? Absolutely not. But it catches a lot of errors, right?

And you're now at the stage where very, very few people will disagree. And some of the people who disagree tend to have vested interests, frankly, behind this. And this has been documented with the fossil fuel industry actually funding disinformation around climate science.

So I think we haven't got to this yet. We have also seen an incredible slew of impacts. This is the warmest year on record by some margin, right?

So we have the models and we have the science. We have the theory. We now have the observations.

I mean, if you look at the graphs of the Antarctic sea ice and its melt, if you look at the graphs of the ocean temperatures, which is the really worrying thing, because as I said, 90% of the heat is in the ocean. Things are way out of whack. And water expands when it's heated, right?

So that's part of the sea level rise story. But the really thing that's really alarming people is the ocean's ability to absorb the excess carbon dioxide now reaching some sort of limit. Which means you're going to see more of it in the air, right?

So let's look at India. Just look globally at what we've seen this year, right? So we had a flood in Libya.

Somewhere between 5,000 to 10,000 people died. It was 200 times the rainfall that is expected for that period. There's now science, which is called attribution science that says how likely would that have been in a non-climate warming world?

And the answer is, it's 50 times more likely in a climate warming world. So we saw that in Libya. We saw wildfires in Canada.

We saw wildfires in Hawaii. We've seen floods in Himachal. We've seen the Yamuna reaching a 45-year peak.

I mean, it's just cascading evidence. So I sort of feel like we kind of have to move on. So what is Navroz's 60-second definition of climate change?

So I'm going to take the science as given for all the reasons I've told you. To me, and this comes back to Bhumi's point, we have to get away from thinking about climate change as a abstraction. Climate change is now a social and political reality that is going to have to change the way in

which human beings organize our economies and our societies, both to reduce emissions as well as to adjust to what are now inevitable harms.

And we need system-level changes for this. It's not marginal changes in how we produce things, in how we consume, really, really important in how we consume, and how we build our, in particular, our cities. Okay.

Who would like to go next? Bhumi, what is climate change?

[Speaker 1] (55:27 - 57:10)

I'll keep it very simple. Our planet's heating up. Everything that Mirek and Navroz said, I completely resonate with it.

2023 was the hottest year in many, many thousands of years. The year before that, 2022, we were at a 1.2, please correct me if I'm wrong, pre-industrial temperature, which is a lot. I think what people fail to understand is that we think 1.4 degrees higher, but that 1.4 degrees is going to have an effect like what we saw in Libya, what we saw in Himachal, what we saw in Australia, in the wildfires, where like over a billion animals were wiped out. There were thousands and thousands of people that lost their homes, livelihoods. There were so many that died. What happened in Hawaii?

You know, the oceans are heating up. We are also going to have mass extinction when our marine species are concerned. You know, I just feel like as a human race, we forget many a times that we share this planet.

You know, it's not ours alone. There are so many other species that we share it with. And I have a very empathetic take on it because I just feel like, for me, climate change is about adapting to newer ways of living.

It is a reality that we are living in today, and it's only going to get more adverse and it's only going to get tougher. I want to be extremely optimistic about it, you know, because that's the tone that, I think that's what the messaging needs to be, that it's not that it's the end of the world, but we have reached a tipping point. And when we speak of 2030, we are literally seven years away from it.

Six years away from it. Yeah, we are literally six years away from it.

[Speaker 2] (57:11 - 57:12)

Sunita, climate change?

[Speaker 1] (57:13 - 57:16)

It's all been said. So, yeah. But for me...

[Speaker 2] (57:16 - 57:41)

But I'll tell you, this is such a curious thing because we asked a whole bunch of people, like a lot of them. And everybody had heard climate change, climate change, but nobody could describe what it is. And it's such an important thing.

Before you start worrying about something, you have to be cognizant to what it is in a way, right? So, we hope we're able to tell a lot of people through this in a very simple way, what is climate change?

[Speaker 1] (57:41 - 1:00:56)

No, it's a very good question because, and we've heard it, I mean, different things. We've got the full picture here. I mean, if I was to wrap this up, I would simply say that, you know, there is something that we take for granted, which is things that we use, largely fuel that we burn, fossil fuels, oil, gas, everything that makes our world so good.

That we have found today is emitting a substance called carbon dioxide, which has a very long life. So unlike, say, the emissions that you have in your city, which are bad, but those are short-lived. So, you know, you have particles, but they, if you clean up, if you can get some wind and if you, and they have a natural life, but carbon dioxide has a very long life.

Now, that is filling up our atmosphere. And it's, as Mirek said, I mean, to me, the most evocative image is that Earth has got a blanket on it. And so that all those gases are being trapped.

And as a result of it, you're getting temperature increase happening. Now, your point about Donald Trump and Vivek Ramaswamy and all the rest in terms of saying, but this is, everything we're seeing is natural variability. The fact is, yes, there is natural variability in weather.

We've always had floods. We've always had droughts. We've always had a landslide, a lightning strike.

But what you are seeing now is the intensity and frequency of those events like never before. And I can only tell you from my personal experience that in 1992 in Rio, when we talked about climate change, we basically knew it was something that would happen in the future. And we knew it was called climate change, but what would happen was something that the models were telling us would happen.

We're living it today. And we're living it today with 1.1 degree centigrade rise post 1850, 1880, okay? 1.1, okay?

If you get to 1.5, just think about it. So we do something called extreme weather database. And we put together all the extreme weather events as defined by IMD.

And then produce a report at the end of each year. It's one extreme weather event a day, a day as defined by IMD. It's not my definition of extreme weather event, extreme rain, lightning strike, flood, drought, all the extreme weather that IMD has, that IMD is the Indian Met Department.

I think that's something everybody knows, okay? But IMD definition, it's one a day in India. Now, if you did the same for the rest of the world, I think even a Donald Trump or a Vivek Ramaswamy would find it very difficult to argue against climate change.

[Speaker 2] (1:00:56 - 1:01:48)

I think before we put carbon dioxide on this villainous thing that it's a bad thing, greenhouse gases are actually very important. Otherwise, life on earth wouldn't have existed. So I think it's the balance part of it.

Exactly. This amount of greenhouse gases, because just to give you some numbers, if there was no greenhouse gas on the same logic, the earth's average temperature would have been 30 degrees lesser than what it is and it practically wouldn't have any life. So I think the context is the right amount for us, for the last 100,000 years that has happened, to go further rather than just saying, okay, it's...

So what is it about the psychology of politicians who are saying climate change is false winning elections? Who are they resonating with and why? Well, it sounds like Sunita has an answer.

I do too. But do you want to go first, Sunita, on this one?

[Speaker 1] (1:01:48 - 1:01:48)

No, no, go ahead.

[Speaker 2] (1:01:49 - 1:01:50)

I'm happy to hear yours.

[Speaker 1] (1:01:50 - 1:01:53)

Okay. So I've been thinking a lot about this.

[Speaker 2] (1:01:53 - 1:01:59)

How do you... Another question while we're talking about this. How do you deal with politicians while you're talking about climate?

[Speaker 1] (1:01:59 - 1:02:57)

I think there are three things happening in the world which is leading to more authoritarian right-

wing governments coming in across the world, which is also why you're becoming more... Denialism on climate change is becoming easier to do. One, I think action is getting harder now.

So the West had easier action, what you call low-hanging fruit. They're all gone. If you look at the United States, so what has the US done to reduce its emissions till now?

The US has gone from coal, the use of coal, to the use of shale gas. Okay. Now most of its energy today comes from shale gas.

And they say that's half the amount of emissions as coal? So that's again the question. So it's half the CO2 emissions, but then when you add to it the methane emissions, which are also greenhouse gas, it's hard to say...

[Speaker 2] (1:02:57 - 1:02:58)

You're talking about fracking, right?

[Speaker 1] (1:02:58 - 1:03:39)

You're talking about fracking, you're talking about shale gas. Basically gas, natural gas. Let's sort of...

But from the technology of it. They've moved to gas. Gas is cleaner than coal, but only relatively.

But they have cleaned up their emissions by moving to gas. Now they have not made any structural changes that we are talking about. They have not weaned themselves out of cars.

Today transport-related emissions are increasing in a big way. They have not weaned themselves out of building big houses. In fact, home-related energy is increasing big time in the U.S. They have not weaned themselves out of eating a lot of food.

[Speaker 2] (1:03:39 - 1:03:42)

And that's the largest emitter, right? Construction industry across the world?

[Speaker 1] (1:03:42 - 1:03:44)

No, transport is today their biggest emission.

[Speaker 2] (1:03:44 - 1:03:45)

Not construction?

[Speaker 1] (1:03:45 - 1:05:35)

No. Worldwide? No.

So nothing structurally has changed. Now, same in Europe. A little better, but not much better.

So they haven't really transformed their... But they needed to. I mean, they had 30 years where they were to reduce their emissions.

They exported it to China. They've really not done much. Now the tough job starts.

And that's when now they're getting a backlash. So that's clearly one reason why the vested interests are kicking in more now. If you look at it, Donald Trump openly said, baby drill would be his biggest slogan when he comes back.

Why? Because Europe needs shale gas. Gas prices are today very lucrative for the U.S. So that is one part of it. There is one other part of it, which is actually an outcome of what you're seeing with climate. The more war you have, you have physical war, which is the war from Syria to Ukraine to Gaza now. You have displacement of people, but you have the war against nature, which is also leading to huge amount of displacement of people.

And today data shows you what they call new displacement, which is not war-related, conflict-related, but flood, drought, agrarian distress-related displacement is huge. Now this migration issue is really hitting and that's also leading to right-wing politics growing up. Then comes the China question, globalization, whether it's paid up in terms of employment.

So it's economic factors that are leading to it.

[Speaker 2] (1:05:35 - 1:05:37)

Are you saying that because like, I'm just trying to simplify.

[Speaker 1] (1:05:37 - 1:05:38)

Yeah.

[Speaker 2] (1:05:38 - 1:05:54)

Right-wing byproduct capitalism, one has a lot, expends a lot of energy and other has lesser versus left-wing, more socialist in nature. Everybody expending lesser amount of energy. Are you making that debate?

[Speaker 1] (1:05:54 - 1:06:18)

No, I'm not making that debate at all. I'm not making right versus left at all. Okay.

I'm making it in terms of the political system, which is much more conducive or much more, which feeds this narrative in terms of there is no climate change and that, you know, we need, they look at it in terms of just hard economic interests.

[Speaker 2] (1:06:19 - 1:08:18)

Geopolitics are so complicated that we can't align a narrative with a certain kind of party. No, no, no, let me sort of, I think we, you know, I agree with Sunita on her characterization of the fact that the economic hurt or the economic adjustment costs, right, both parties in the US, all parties in Europe, including incidentally in Germany, right? I mean, there's been a change in part over the last decade, but there has been very little political appetite to bear any of the costs of adjustment to climate change anywhere in the world, right?

Where I think it gets interesting though, is we are at a moment now where we're teetering on the edge of an opportunity story where countries are now seeing their future economic gains actually being about building low carbon economies, partly because China has spent this last decade very assiduously building these industries and suddenly the West is playing catch up, but they don't, but the question now has become the speed of the transition. If you can't do it fast enough and you take on obligations, then you still impose costs on the economy and we are globally in a bit of a catch 22. So you now have a situation where the West is really retrenching.

China and India is trying to get competitive in this green economy. It takes a lot of money. It takes a lot of ability to throw massive subsidies at it.

The big question for me is, India might be able to keep pace and we can talk about this later, but what happens to Africa? What happens to the smaller countries? They're going to be basically getting the crumbs from this table of green industrial subsidies.

So we might be in this dilemma where the most aggressive things to solve climate change might actually make for a more unfair world. That's the kind of tension we are facing.

[Speaker 1] (1:08:18 - 1:08:19)

It doesn't have to be.

[Speaker 2] (1:08:20 - 1:08:58)

But to the point, like the one sort of bright spot in this whole thing is you mentioned economics, how economics will in the end drive. So that's where if economics of climate change solutions, whether it's to some subsidies in the front, they just end up becoming better. The world needs any sort of growth rate in the terms that people want to use the world.

You have a growth rate when you're transitioning away from all these things into something new. You have to build new assets and things like that. So once there's an economic alignment that it just makes sense with the fact that you need to transform all these assets, I think as you rightly mentioned, that will take over the narrative.

[Speaker 1] (1:09:00 - 1:10:06)

There's even better. I mean, the fact is we haven't built the assets yet. So our big advantage, Africa's big advantage is that we can build them differently and build them.

That's where the huge investment story should be happening. I mean, frankly, but it's not. I mean, that's what we need to look at.

Why is money not going to the countries that need it most, who have the biggest opportunity to reinvent? And I think if we can understand that and fix that, you actually can change this complete landscape. The West has a problem.

Let's put that aside for a moment. It has a problem, we'll deal with it. But we have an opportunity.

Now, today, if you look at the investment that is happening, and this is the problem, the money that is coming in the name of climate finance is actually not going to anywhere in Africa, not even coming to some extent in India, largely going to China and a few other countries. It's not going where it is needed most. Why is it not going there?

Because the cost of capital is too high.

[Speaker 2] (1:10:06 - 1:10:14)

There is a lot of talk around the Bridgetown initiative and things like that around transforming the multilateral development banks and the World Bank.

[Speaker 1] (1:10:14 - 1:10:40)

Exactly. But what I find, two things that have happened in this COP, which one, we have a clear understanding that it's not the amount of money, but the quality of money. So in fact, there is a lot more talk about how you need to reform the way the money, who gets the money?

How does the money come? Does it come with an interest loan? I mean, I'll give you a simple example.

So you heard about, we all know about the floods in Pakistan.

[Speaker 3] (1:10:40 - 1:10:40)

Yeah. Okay.

[Speaker 1] (1:10:41 - 1:11:08)

Everybody cried about it. Huge chest beating, you know, Pakistan in trouble. Huge conference was created to discuss Pakistan and money was given to Pakistan.

You know how the money was given? It was given as loans. Now in a country, which is already

under so much debt, giving it a loan to fix its flood problem only means that it gets further indebted.

[Speaker 2] (1:11:08 - 1:11:13)

But all the 100 billion by developed countries, I think they are planning to do it from the loans itself.

[Speaker 1] (1:11:13 - 1:11:15)

So that's 75% is loans.

[Speaker 2] (1:11:15 - 1:11:15)

Okay.

[Speaker 1] (1:11:15 - 1:11:17)

So it's more or less loans.

[Speaker 2] (1:11:17 - 1:11:18)

And World Bank will be the main institution.

[Speaker 1] (1:11:18 - 1:11:51)

So, but no, but that's where the conversation this time with the Bridgetown initiative, with many others, talking about low financing and the other issue that came out this time, which I, you know, is the need, because private sector is not investing enough in climate change. And the question is, how do you get them to invest? And there, the big issue came out is, well, they will invest if there is carbon credits and offsets.

And the big problem there is the quality of that investment.

[Speaker 2] (1:11:51 - 1:11:55)

The big issue last year with the very bad offsets.

[Speaker 1] (1:11:55 - 1:12:00)

Yeah, we did a detailed expose and we were not liked by people at all.

[Speaker 2] (1:12:00 - 1:12:44)

We should absolutely come back to that whole question. I just want to make the observation and picking up on what Merik said earlier and tied to what Sunita also said, you know, it is going to be very hard to bring all these pieces together. But there is an interesting story here where low

carbon futures are now seen as inevitable and an opportunity.

That tipping point that I used to hear about all the time. Low carbon futures? As in the path to being competitive for an economy is to actually be ahead on technologies and their deployments that make a low carbon economy.

[Speaker 1] (1:12:44 - 1:13:09)

Take the iron and steel sector. Sorry, just following what you're saying. Take the iron and steel sector.

We need to double our iron and steel production. There is no question about it. We're very low.

We need to double. But there are ways of doubling that without emissions. So you could be at lesser emissions than you are today.

Then at the point you're doubled.

[Speaker 2] (1:13:09 - 1:13:48)

So whoever gets there first will have a competitive advantage, right? That has become the norm. And so we've seen battery costs.

So our solar panel costs fell 80% during this last decade, right? Battery costs fell considerably as well. Wind costs fell 50%.

And so now you're in a world where it's going to be difficult to adjust, but it's possible. And the key thing is there is an incentive for everybody to adjust. So to that point, I think you mentioned about batteries cost.

I think one important cost that has to be solved this decade and it's again a point to the entrepreneurs out there is around grid storage.

[Speaker 3] (1:13:48 - 1:13:49)

Grid storage.

[Speaker 2] (1:13:50 - 1:14:33)

Because your solar and wind can't really scale after a point if you don't have very, very good grid storage. And I think those cost curves, because I don't think lithium-ion can really work in the grid for whatever its challenges are. And lithium-ion always will get a priority to the vehicles.

And again, there's a supply-demand deficit there already. Grid storage technologies, again, to your point of what can entrepreneurs do? I genuinely feel grid storage and innovation around

that, that whole supply chain, non-lithium-ion base is an exciting opportunity because the amount of grid storage that will be required.

I think India's number is, if I'm not wrong, it's around 200 gigawatt hours by 2030. And we have not even stabilized the grid.

[Speaker 1] (1:14:33 - 1:15:41)

But Mirek, if I may take this conversation forward and ask you as well on this. So there are two problems we are handling right now. One is, so we're all passionate about decentralized solutions.

And as you saw, my fecal sludge, my day-in-water harvesting, all that sort of really excites me to see putting things in the hands of people rather than in the hands of government. So it's exciting. Build a lake, a pond, put it in the hands of village communities to run it.

That's the resource management. So many years ago, we started working on mini-grids. Okay, very passionate about it.

Went and saw some mini-grids in Chhattisgarh, other places. So I keep coming back to it because I don't want to leave it for our research and work. But the cost is just so high.

I mean, at 30 rupees a unit, it's not possible to basically replace. Now, either you had a feed-in tariff situation where the government would subsidize it. But, you know, it's huge.

And that's the scaling issue.

[Speaker 2] (1:15:41 - 1:15:56)

That's what the point is that this is the decade where energy storage has to... It has to happen to energy storage what happened to solar in the last decade. But once you hit that tipping point, in a way, of a solution, then suddenly you're off-grid.

[Speaker 1] (1:15:56 - 1:17:05)

So my question is, and I'm asking myself, these questions I'm asking you. So we also set up, we have a large facility where we train people. And so we want to practice everything before we preach.

So my thing is that we do our own shit management there. We do all our stuff there so that we can preach properly. But so we set up a solar roof top system, large one too, and we thought we'd sort of go.

Now, the problem that's coming is that the battery storage costs are too high. So I can't go off-grid. So my DISCOM is my backup, my customer.

They're also my backup. So nighttime, I buy from them. Daytime, I export to them.

Ideally, that should work very well. But the problem is our DISCOMs are... I mean, to me, even as a proponent of mini grids and really arguing for it, I'm actually getting more and more convinced that if you need to scale up, then we have to.

And you have worked more on this than anyone I know is the question of DISCOM reform. And it's not happening, Navroze. I can tell you from the little state...

[Speaker 2] (1:17:05 - 1:17:07)

If you had to summarize the problem with...

[Speaker 1] (1:17:08 - 1:17:11)

Let Navroze do it. He's the one who spent most of his life looking at it.

[Speaker 2] (1:17:11 - 1:17:13)

Like very simple definition that everybody can understand.

[Speaker 1] (1:17:14 - 1:17:19)

What is wrong with India's power grid? That I will give you, Navroze will give you a complicated. That's between him and me.

[Speaker 2] (1:17:19 - 1:17:22)

That's why I'm nudging him simpler. That's the role we play.

[Speaker 1] (1:17:23 - 1:17:25)

He complicates it, I simplify it.

[Speaker 2] (1:17:25 - 1:17:44)

Well, the good thing is I haven't worked on it for a bit, but the good thing is and the bad thing is that it hasn't changed much, right? I will be very straightforward about this. The issue is that distribution...

DISCOMs are distribution companies, right? And they have been...

[Speaker 5] (1:17:44 - 1:17:45)

Elaborate for 30 seconds.

[Speaker 2] (1:17:45 - 1:18:39)

So the electricity sector has generation, transmission and distribution. And in India, we set this up as originally tying all of these things together at the state level. And then through a series of various changes and reforms, the distribution companies were kind of hived off.

Just like breaking into that. Yeah. When you say state level.

Yeah. Electricity generation is not state level. No.

So you're saying distribution is state level. That it is now. It used to be the case that in each state, there was a what was called a vertically integrated generation, transmission, distribution, all together in a state, right?

The breaking apart was in a sense to try and create markets and generation, which didn't work terribly well. It worked in part. Markets and generation meaning?

As in to allow private sector to invest in generation companies.

[Speaker 1] (1:18:39 - 1:18:40)

But it's happening.

[Speaker 2] (1:18:40 - 1:19:23)

Which is happening. And this was the era, the most infamous part of this was the era of Enron, right? Enron came into Maharashtra in the 1990s and so on and so forth, right?

And so there were complications around bringing in private providers. So this, what this goes back to is to bring in companies like Enron. They said, listen, why would we invest unless we're sure we're going to get paid for electricity?

So they were given all kinds of guarantees and incentives and so on and so forth. Because they were an energy-scarce company. Enron came into India.

Enron and several other companies. Started generating electricity through coal? Through coal.

Okay. And started selling to a particular state. Which state?

Well, they said... Maharashtra.

[Speaker 3] (1:19:23 - 1:19:23)

Maharashtra.

[Speaker 2] (1:19:24 - 1:21:56)

But they basically said that we need to have all kinds of guarantees before we come in. Because

they didn't have storage of electricity. They had to sell at the time they generate.

And back then we weren't talking in terms of storage. And electricity is kind of a gigantic machine, right? It's not a commodity you can store.

It has to, as it's generated, it has to be used or stored. Now, what ended up happening is in order to attract these companies, you had to give all these guarantees, which then put a lot of stress on the distribution companies. Now, the distribution companies were...

Many of them were actually operating in the red, right? They weren't making money. And there's a reason for that as well, right?

The reason for that is that politically it became very convenient to win votes. To basically tell farmers and low-income groups, look, we'll give you discounted prices, right? And there's nothing wrong, a priori, with giving cheaper electricity to low-income consumers.

It's done throughout the world. What happened in India is that they said, we'll give free power to farmers. The farmers stopped being metered, right?

And when the farmers stopped being metered, you didn't know what was happening to the electricity. And so industry started conniving with the linesmen to steal electricity and saying, oh, it must be the farmers. So what started happening is there was this accounting hole in the middle of the distribution companies.

And nobody quite knew who was stealing the electricity and where it was going. They stopped becoming functional entities. And the challenge of making them viable and imposing discipline on them has been on for 20 years.

So we now have... It's actually kind of amusing, right? We have the accelerated APDRP, which is the Accelerated Power Sector Reform and Development Program.

We had APDRP-1, APDRP-2, APDRP-3. Then we had Uday-1, Uday-2. Now we have a new one.

What is APDRP? What is it? So it's all basically various efforts to try and discipline the distribution companies and say, if you tell the state governments, if you make sure that your distribution companies collect money and pay the generation companies, we will give you certain carrots.

And if you don't, we will impose certain sticks, right? The problem is that you are treating this as a technical problem. It's not a technical problem.

It's a political problem. Has it changed now? It's not really changed.

It's better. So right now Adani, Ambani, they produce power. They produce power.

[Speaker 5] (1:21:57 - 1:21:57)

Tata.

[Speaker 2] (1:21:57 - 1:22:00)

And they sell in Maharashtra. A lot. Everywhere.

[Speaker 1] (1:22:01 - 1:22:03)

They sell actually to the national grid now.

[Speaker 2] (1:22:03 - 1:22:48)

So what you have to be careful of, it's assuming that this is the same story in every state. In every state, the problem is slightly different, right? Can I just for magnitude ask you, from power generation to power utilization, just for scale, how much power is lost through pilferage, through inefficiencies, all of that as a percentage?

So again, it really varies by state. The technical losses should be kept at around 5% to 7%. We now have this measure called aggregate technical and commercial losses.

Commercial losses, i.e. theft. And that can vary from 15% to 30% across states. In the worst states.

I think it's come down now. I'm not completely up. It wasn't even without the kinds of theft.

[Speaker 1] (1:22:48 - 1:23:16)

I think one of the points we miss when we talk about this, and I think that brings us back to the energy, the renewable energy question, is the fact that we produce energy from the dirtiest possible source possible, which is coal. It's also the cheapest source possible. At the point of production of that energy, it's about a rupee, two rupees, even less.

It's actually, it could even be less than...

[Speaker 5] (1:23:16 - 1:23:17)

It's about two rupees.

[Speaker 1] (1:23:17 - 1:23:40)

Now it's two rupees because of the other costs. But a decade ago, it was even less than that. By the time it gets supplied to a customer, it's about seven to 11 rupees.

So the first question that we have is the inefficiencies in the system that take the supply from the

one or two rupees to the 11 rupees. That's one.

[Speaker 2] (1:23:40 - 1:23:57)

Can I ask who is generating it at two? But just to clarify though, the two rupees is the, it's a two part tariff, right? So there's a capital cost and there's the fuel cost.

Two rupees is the variable cost. And then on top of that is the... So if you add up both of them...

[Speaker 1] (1:23:57 - 1:24:03)

If I take Badarpur and I take some of the older power plants where the capex costs are pretty low, it could be two rupees.

[Speaker 2] (1:24:03 - 1:24:52)

Then because we depreciate, depreciation will happen. That's right. So the average is now 3.6 for the average cost. Including capex. After depreciation, a new plant is closer to five. Okay.

This is for coal. This is for coal. Just before we get into the next part, what is it for solar, wind and...

Right. So for solar, the lowest you can get these days is around two rupees, right? Now it gets complicated because they're starting to say, well, what about solar and storage?

So solar and storage ends up being around four rupees, but it's not enough storage to come back to the... The original exchange all had, it's not enough to replicate coal. What about people talking about things like solar in the day, wind at night, solar converted to hydro and stored?

So it's more...

[Speaker 4] (1:24:52 - 1:24:52)

That will be storage.

[Speaker 2] (1:24:52 - 1:26:28)

So that will be a pile of storage. So what is the eventual cost of this come down to? Let's say I have a solar plant.

I'm converting it into hydro and I'm disseminating via that. Hydrogen or pumped hydro? Pumped hydro.

Compare that to coal. What is the difference today? I don't know the...

I know that lithium is pretty high. Pumped hydro may be fine. I think there's a prior point.

The way in which we have been talking about this, because you've had power systems engineers that have been trained for a long time, is you're thinking about renewables and you're saying, how do we make it look like coal? That might actually be the wrong question to be asking. Right.

And so you have folks who are working on decentralized power who... And this is the mindset shift that I'm talking about. Why is it not just about technology?

It's about technology. It's about institutions like the DISCOMS and it's about the politics, like free power to farmers. What if you turn this around and said, what we actually care about is not energy.

We're certainly not about coal or renewables. It's not even energy. It's about the services that the energy delivers.

Then you can think much more flexibly. So for example, one of the really innovative programs that we're trying to get to take off in India is can we move to solar for agricultural pump sets? Because you don't need 24-hour power.

Farmers are irrigating during the day. And maybe not from a grid, but at the farm. And what's the genius of this?

You take that demand, you take it off the grid, basically. And this big accounting hole that I talked about... Is the CAPEX cost prohibitive?

Is it too expensive?

[Speaker 1] (1:26:29 - 1:27:03)

There's a huge subsidy for it. The problem there is the level, where is the groundwater? How deep is it?

So that's why it's an intersecting problem. I mean, that's where you need now... You need to combine it with two other things.

You need to combine it with ability to harvest your rain, to be able to get your groundwater back into shape. And you need to combine it with cropping patterns, which don't overuse your groundwater. So you move away from...

You can't have a nice world of sugar cane, and use a solar panel and tell me that it is clean.

[Speaker 2] (1:27:04 - 1:27:16)

Can I summarize and say that we defined what is climate change? Now we are saying that the biggest solution around this is energy transition? It's definitely the single biggest.

[Speaker 1] (1:27:17 - 1:27:23)

Energy is at the core of your climate change crisis and opportunity.

[Speaker 2] (1:27:24 - 1:32:06)

Specifically in India, you're talking to the Indian government. That's right. One thing they can do to make the power grids of India better?

They need to change and mandate that the distribution companies become the agents that provide this backup power and organize their own power procurement for that purpose. So it's a little complicated because it's about the nature of power planning. So power planning is done today for centralized systems.

We need power planning for decentralized systems. And it's a different form of power planning. If I can ask you and it's a simple...

I don't know whether it exists. Like if you just do time of day pricing, like differential pricing, then like that's the simple thing. Like if you just have differential pricing based on the generation, then automatically the demand shifting also may start happening because then if you're in the night...

So I think for me that can be one... It doesn't happen right now. No, so time of day pricing is one important thing.

Would you say that's the easiest low-hanging fruit? It is definitely one of the things to put in place. I don't know whether we know enough to know exactly which is the best lever to pull.

We have to pull three or four. But I know we have to change the incentives for the distribution companies. Right now, the distribution companies see renewable energy by individuals or by small entities as actually a threat to them.

But the second thing I think is really interesting and gets to the core problem I told you about earlier, right? Which is when... Which is providing electricity...

We're in a loop right now of providing bad quality electricity to poor people who say, well, why on earth should I pay? I don't even get electricity when I really need it. And because they don't pay, then they get even less service, right?

We need to be actually invoking a very different conversation around electricity, which is not electricity for the sake of electricity, but electricity as a service. And this goes all the way back to somebody Sunita will know, a guy called Dr. Amulya Reddy, who's sort of the father of this way of thinking about electricity as a service. So the way we've been thinking about in our organization is, what have you talked about not providing electricity for the sake of electrons, but you're providing ways of increasing the productive capabilities of people in rural areas?

So you're thinking about not subsidizing consumption of electricity, but you subsidize the things that make electricity productive, like coal chains, food processing, small industry, you subsidize that. And then you don't have to subsidize the electricity itself. And then people have an incentive to actually pay for it, right?

Because if it's not good quality electricity, they can't make money. You can also bridge that with a renewable energy story and have intermediate companies coming in and saying, we will finance and find ways of bringing in this product and we'll sort of amortize it so you can pay us over time. So the idea of providing not electricity, but enhancing productivity from electricity.

All of what you're saying is sounding very simple. Why isn't it happening? If it was that simple?

Well, I think partly it does take just a question of re-imagining and partly you have institutions, in particular the distribution companies, who see their incentive is in selling electrons and in not selling electrons to people who are loss-making. That's the dynamic we're in right now, right? If you're going to lose money for every electron you sell, you're selling it at one rupee, it costs you 356, you're going to find all kinds of ways not to sell it.

So shifting the incentive structure is really important. Okay, moving on. Next topic, carbon concentration in the atmosphere.

1800s, it used to be 290 ppm. They say the world is okay at 350, right now we're at 420. There is some evidence to say that higher percentage of carbon, higher the temperature.

Now, 1.1, 1.2 degree is the rise so far recorded and everybody's talking about where we will be by 2050. 1.5 to 2 seems okay. Anything higher sounds crazy.

Is that broadly right? 1.5 is okay? It's okay in a sense of?

1.5 is doable.

[Speaker 1] (1:32:07 - 1:32:09)

1.5 is doable meaning?

[Speaker 2] (1:32:10 - 1:32:36)

And by the way, it's 2100, I think. The average is 2100 or 2050, they are saying. I think the sense is 2.6 is sort of the, 2.6 to 3 is the business as usual scenario. Business as usual. 2100, I think everything is 2100. In terms of what we're doing now, right now we're on track for somewhere around 2.6. Paint me the dystopian picture of 20% of the world dying, what temperature increases that?

[Speaker 1] (1:32:38 - 1:32:41)

I don't think that's a question that anyone can answer.

[Speaker 2] (1:32:42 - 1:32:42)

Not with any certainty.

[Speaker 1] (1:32:43 - 1:33:49)

Quite frankly, that's also been one of the problems and that's why you're getting a lot of people saying climate change doesn't happen. Climate change is not about that image that you have, that sort of very cinema image that you have, that you're going to get these vast numbers of people just dying and just... I mean, we saw that in COVID, okay?

So we have lived through it. We have seen COVID happen where we've had exactly as you did Bhoomi, calling up oxygen cylinder. I mean, so many people we know died, okay?

Because of, you know, what? But climate change is not that. And I think we shouldn't paint that image in our head because that tends to get a lot of people saying, but that's not happened.

So that means climate change is not going to happen. I think what climate change is about is about the fact that you're going to get nature, which is on a whack, it's out of control, okay? You're going to get more and more extreme weather events.

You're going to get more and more deviations between what's normal and what's not normal.

[Speaker 2] (1:33:49 - 1:34:22)

Bhoomi, can I ask you a question? Like Hollywood back in the day has made smoking cool, has made diamonds expensive, has defined what we perceive to be attractive today beyond symmetry, if one might say. Why doesn't Bollywood, your industry?

You've done so much work, right? Social causes, movies around that. Don't you think it's a good way to reach a large plethora of people by making movies around climate?

[Speaker 1] (1:34:22 - 1:36:22)

Absolutely, but I feel it comes from the fact that a lot of people feel they're not going to be affected by it. You know, because we feel that because we are protected by the wealth that we have, there will be some elite institution that we believe in that is going to come and solve this. But that's not the case.

As ma'am said, you know, as she rightly said, that climate change is something that is going to be a great equalizer and there'll only be enough narrative around it, which I really feel is missing, especially in the industry that I am from, because people feel it's not going to affect them. You know, we live in a world, we live in a bubble and our bubble gives us a lot of comfort, right? We

are still, we have access to the best of things.

I feel the narrative will only shift when we are truly affected by it. And that's what happens. When we started our conversation, that's exactly what I said, that even today when I am in rooms and this conversation has been so educational for me because I'm just taking in so many different perspectives and I'm not privy to information like so, right?

What my path in the fight is that I'm just trying to get as many people like me to start talking about it. You know, when it needs to become popular culture, that's the impact that media or films have, right? I want climate change and all the effects and I need people to be fearful in many ways.

And I really hope that there are many more films that are made. In fact, there was a film called Karvi Hawa that was made. It was such a phenomenal film, but a film like so didn't really, it didn't really do well.

It didn't reach a large audience because it also makes people feel they rather have an ostrich mentality where they behave and they pretend like everything is okay around us. You know, someone will come and save us from this situation. But that's not going to happen.

[Speaker 2] (1:36:22 - 1:36:26)

Do you think fear sells more or does greed when it comes to movies?

[Speaker 1] (1:36:27 - 1:36:39)

Greed. Greed. Any day greed.

Absolutely. I mean, how many of us are going to go sit in a theatre, kind of go away from all our issues and our problems and be like, oh, this is a real issue.

[Speaker 2] (1:36:40 - 1:36:47)

So it might work for climate change in movies. If you paint a picture like Armageddon or, you know, these end of world stories.

[Speaker 1] (1:36:48 - 1:37:27)

Look, my generation has seen these films, right? But we still, we know it's happening. We believe in it.

But how many of us take action again about it? None of us take action about it. My industry is a polluter.

You know, like if you see film sets, film sets are not run in the most sustainable manner. Wherever we go, we actually destroy the ecosystem that we are at. I have shot in like remote

villages.

Yes, we are creating job opportunities, which does happen when we go for those three months or six months. But we leave behind, we disrupt the place. We pollute it in so many ways and we just move on.

[Speaker 2] (1:37:27 - 1:37:30)

Is that because there's no financial incentive to do otherwise?

[Speaker 1] (1:37:31 - 1:37:37)

Maybe, but also I feel like... Oh no, debtors. Nobody holds them to account.

[Speaker 2] (1:37:39 - 1:37:44)

That's the question I'm asking myself. Is it financial incentive or debtors that will work?

[Speaker 1] (1:37:44 - 1:37:52)

I feel financial incentive would definitely work the moment people realize that there is some way that maybe we can... They will cheat around it.

[Speaker 4] (1:37:53 - 1:37:56)

We have a long history of...

[Speaker 1] (1:37:56 - 1:38:01)

Let's do some nice cover up. Beautiful, you know. We painted it green, see.

[Speaker 2] (1:38:02 - 1:38:06)

We have a long history of not being able to make financial debtors stick.

[Speaker 1] (1:38:06 - 1:38:16)

We need both. What is the solution, right? How do you get people to actually care and bring about a change?

You need society.

[Speaker 2] (1:38:17 - 1:38:56)

It's also about what are you trying to project, right? So the movies around climate change, the apocalyptic movies. I don't know if you guys remember this Don't Look Up movie climate change from an asteroid.

It's very destructive actually because that's not the way it works. It's not that Armageddon as Sunita said. Imagine it is that Armageddon and when they see in reality it is not the disconnect.

I was thinking like this but it's... Should you make movies about floods in Bangladesh? You have to make it about the lived experience.

What happens when you have a heat wave? What happens when you have a flood? So climate change is the cause behind the scenes.

But you don't make movies about climate change.

[Speaker 1] (1:38:56 - 1:40:05)

Having said that there was a film that I was a part of actually that was about what happened during COVID, right? People just didn't have the appetite to go watch that film. You know because it is their harsh reality and they don't want to live it.

Very good point. So how do you get the masses to kind of accept what's happening and to bring about change? Like I work on a very micro level, right?

Like on my platform I'm propagating, okay how did I make the switch to living a sustainable life? You know what are the smaller things that you can do? I don't think people changing on an individualistic level is the solution because I think the solution comes on a policy level exactly what all of you were speaking about.

But if there is a mass shift in the way people think... For example, I did a film that was associated with the Swachh Bharat Abhiyan which was about open defecation called Toilet Ek Prem Katha and we did see a certain shift in thinking patterns. We saw a behavioral shift, right?

It kind of ties down to your experience of...

[Speaker 3] (1:40:05 - 1:40:06)

It's a very good movie.

[Speaker 1] (1:40:06 - 1:41:30)

No, it's a very good and remember the toilet story till then was about building the toilet but not using it, right? It was films like that which actually made people understand the cultural and the social reasons why you would adopt toilets. So, you know, I think you've really...

So, Bhoomi, it is possible to nudge people. It is possible to nudge societies and change the way they think into doing things differently. But again, with the toilet story, there was both.

There was a cultural change. There was also a subsidy. There was an incentive as well given by

government, big incentive given to build those.

And then there was a reputational advantage that, you know, I mean, if I'm modern, I'm going to use a toilet. So I think there was a lot of that which we need to learn from and bring into many of the thinking issues that we are talking about today because we need to deconstruct many of these conversations because when you talk about this big thing, it looks so impossible. You need to bring it down to...

So I can see of, you know, I also sometimes feel this sort of 20 ways to save the earth is not going to happen, what I can do. But we shouldn't disable people's ability to be able to be different and to do it in their own way differently, practice the change.

[Speaker 2] (1:41:31 - 1:41:57)

But how does it matter if all life is transient? If the people you bring into this... And I have like some really close experience with this.

I had like this realization, I had some drastic personal event happen in the last 20 days and it just got me thinking that everything is so meaningless. We're all born, we die. We're not so much different from any other animal out there.

[Speaker 1] (1:41:57 - 1:42:23)

That's why if life is so transient, we need to do our best to make it work. I mean, I had a cycle accident near death. I mean, I have metal plates here.

So I was on a cycle and the car rammed into me. I mean, near death experience. I mean, all that only made me realize that you just cannot give up.

And I think that should be the way. I mean, I would not... I'll leave it to you to answer for me.

[Speaker 2] (1:42:24 - 1:42:27)

So you're making my argument. You're saying she should not think...

[Speaker 1] (1:42:27 - 1:42:34)

No, she shouldn't think about it. She should use her anxiety to make sure that we can get things done. And that's exactly what I'm trying to do.

[Speaker 2] (1:42:34 - 1:42:40)

Exactly. Do you think this is the world countering the problem of climate change on its own?

[Speaker 1] (1:42:40 - 1:42:40)

No.

[Speaker 2] (1:42:41 - 1:42:58)

Because in a hundred years, we might have less people than there are right now. Because we're talking... That will anyways happen.

Because as economically development happens, that the population peak, I think they are saying it's around 2060 at 10 billion. So by 2100, your number is not going to be that. Any case...

And what happens by 2150?

[Speaker 1] (1:42:58 - 1:43:25)

But the point is, what are you going to be? An American or an Indian? I mean, a poor Indian.

If everybody who is in the world, let's say the population of the world peaks and the population starts declining. But everybody consumes like an American, you're not going to save climate change. If you really...

It's not the numbers. It is numbers plus consumption. And that's the tough one.

That's really the tough one.

[Speaker 2] (1:43:25 - 1:43:56)

So I don't mean to at all diminish the sense of anxiety. But I think it comes back... I'm a great believer in narratives.

It's a big part of what I've tried to work on all my life. Coming back to this question of movies and what you... If you make a don't look up kind of movie, it's going to breed anxiety.

But if you turn this around and you say, yes, this is really an issue. There's a part of this, which is the low carbon energy transition. But there's also a part of this that says, we know we're already in territory of serious climate harm.

[Speaker 5] (1:43:56 - 1:43:57)

Who would watch that movie?

[Speaker 2] (1:43:57 - 1:45:21)

What can we do? No, so I'm just coming to that, right? What can we do to fix it?

Do you really think anybody will watch that movie? I think they will. So for example, the Toilet Ek Prem Katha story was because it was not abstract.

It was very real. And it was couched in a story around how you change positively, right? What the solution is.

We've been working, for example, on heat waves. There are... Heat waves are going to happen.

They're going to happen much more frequently. The impact on people's lives as a result of those heat waves can be ameliorated, can be reduced substantially by community action, communities organizing with the support of the states. If you had a story that unpacked this idea of heat waves and what you do about it in a world where we're going to see more of them, that's an interesting story.

It's a counter to anxiety. Somewhere I ever thought like, like what Shark Tank did, right? So for education to the masses about this whole VC ecosystem and everything is really like, even after passing out IIT with so many startups around, I had no idea what a VC was in a ways.

And now like I interacted with 600 students and they are like, how do I increase my valuation? Which is a very interesting question. But the point is, maybe you don't need to do it outside what is there.

Like I always imagine, like a Tarak Mehta ka ooltah chasma having an episode focused on heat. Right. So you may...

[Speaker 1] (1:45:21 - 1:45:25)

Like The Simpsons have done it, right? Like through and through. That's what they did.

[Speaker 2] (1:45:25 - 1:45:29)

So you may not need to create some new intellectual property.

[Speaker 3] (1:45:29 - 1:45:31)

The main idea can be...

[Speaker 2] (1:45:31 - 1:46:16)

Like one of the things that I feel that people should look at is in all their fields, they should look at, can I propagate climate while not part of that field itself. You don't need to go completely outside because then it'll be too out of the box. So that's my sense.

Like I always feel... Or an IPL. So just from an awareness perspective, I think the EPL had a net zero match.

And I don't know if Tottenham versus Chelsea or something. Now imagine an IPL final with the broadcaster also involved and talking about this is a net zero match with the specifics and things

like that. I think that does create some momentum.

But it has to be maybe involvement across different layers. Yeah. The players also talking about that or all their clothes for that match made from...

[Speaker 1] (1:46:16 - 1:47:01)

And the fact is we have to make sure this is real. Because my other problem, Eric, is that we are getting too... We're getting too glib about this whole thing.

It's very, you know, because I'm a climate change activist, because climate change is important. So then everybody is into net zero. Everything is net zero.

Everything is carbon neutral. Means that I can do everything I want, but I have not somehow emitted or I bought emissions. It's not adding up.

And I think that... I mean, I'm only saying this because, you know, we just don't have time to lose anymore in anything which is glib and doesn't add up. And we don't have time to waste.

[Speaker 2] (1:47:01 - 1:47:31)

No, I just want to query you, Sunita. I mean, I completely agree with you in the world that we work in the most, right? There's the mitigation side of things.

And there's this... Being this glibness and this like, you know, the narrative is caught on. And basically what it means is that more and more people and companies are trying to show that they are acting without doing anything fundamentally different on the ground, right?

I think on adaptation, it has the potential to be a little bit different because it automatically becomes much more real.

[Speaker 1] (1:47:31 - 1:49:01)

But keep the mitigation story also. I think there's a lot of real stuff we could do. Absolutely.

There's a lot of real stuff that companies can do, that people can do. I mean, there are lots of things can be done that can reduce emissions. Maybe you will not have the same wealth, but you would have well-being.

And there is... We need to understand that it's possible. My problem is that we don't tend to do it and we tend to be able to get away with it.

And that's where I feel that one of the biggest roles that we are missing in the climate change is the ability to be able to call out a spade a spade. I mean, when Greta started her movement, I think the shock value of having a young person say so, so clearly, so loudly, and with so much

conviction that, you know, this is just not adding up. Unfortunately, she still remains incredible, but there is now a growing movement of not you, but I can see you having done something and therefore there's a realness that I can see.

But I see a lot of young people who are just getting co-opted by the system, by which they're part of the system. They go there, they say the things, but it's not adding up. So I'm just saying, my advice just simply is that I think we cannot afford to lose another decade in sweet nothingness.

[Speaker 2] (1:49:01 - 1:49:23)

Yeah, no, but Sunita, to the point you were making of co-opting, I agree with you a hundred percent. Somebody wise once said, don't fear the man who has a hundred books in a library, fear the person who has one book and has never read it. And I feel like society at large is like that today.

Unfortunately, this ecosystem is also filled with a lot of people like that.

[Speaker 1] (1:49:23 - 1:50:47)

But I think we have enough people who are not like that. And I think to me, when I think back, I think the fact is that we need to give those people more voice, more space, because otherwise you end up with just being able to get away by believing that you're making a difference, but you're really not making a difference. And I believe a difference can be made.

It's not as if I've opted out of the system and said, nothing is possible, nothing. I mean, I'll give you a simple example. We just did this study called discredited, looking at the voluntary carbon market.

Now, I very strongly believe in the role of markets and carbon crediting to make transformation. Why? Because you need money and you need to get large amount of money for the kind of technology and other needs that we have.

Now, if you could mobilize the market to say, okay, I am interested in getting an offset. An offset simply means that I emit, but if you reduce your emission, then I will buy that emission reduction from you, put it in my account, and I take the credit from it. Do you think the means of offsetting is accurate?

So, you know, when I study, it's a scam. It's not even a, it's such a fraud.

[Speaker 2] (1:50:47 - 1:50:58)

We tried to invest in it. The first company we found was somebody selling TOWs, which are more efficient, the large Indian company. And they were playing that off to be an offset.

[Speaker 1] (1:50:59 - 1:51:00)

But can I say?

[Speaker 2] (1:51:00 - 1:51:00)

It's global.

[Speaker 1] (1:51:01 - 1:51:04)

It's globally not. But it's, yeah, it's not just an India story.

[Speaker 2] (1:51:05 - 1:51:10)

This is the shock of the industry, which is happening. I am hopeful that it will at least mature in the down...

[Speaker 1] (1:51:10 - 1:51:35)

I'd be very surprised. Unless you have rules. Unless you have integrity, you have regulation.

I mean, the industry today is not doing, I mean, they're basically saying good things. I mean, they're saying we're investing in a smokeless chulha. So giving you a chulha, which uses less wood, therefore reduces the amount of wood that you use.

We are investing in trees that are planted. All good ideas.

[Speaker 2] (1:51:35 - 1:51:41)

So if I have X amount of money to allocate towards climate change, is it best spent on trees or something else?

[Speaker 1] (1:51:41 - 1:52:33)

No, no, a lot of people have been buying, a lot of people. So if you look at the offset programs, if you look at the analysis we did, so Michelin, Danone, a lot of the companies that are claiming carbon credits are claiming it on the basis of the trees that they have bought the offsets from, the carbon credits from. The problem is, let's, you know, that's, this is the...

The problem is that cost of the tree on whose land it's being planted, whether the person is benefiting from it, those are not the questions, okay? You want a cheap deal. You want a cheap emission reduction.

You want to basically say, I can continue to pollute, but I have bought a carbon credit from wherever, okay? And I have paid pittance for it.

[Speaker 2] (1:52:34 - 1:58:36)

The price of the carbon credits is like few dollars, under a hundred dollars in some cases, yeah. So let me just summarize where we are, because this is quite long. We started with identifying what climate change is.

We then went into energy transition being the real solution. We spoke about solar. We spoke about wind.

We spoke about what's wrong with the grid. We're debating fission fusion now and the likelihood of it working. Next, we will move on to things like mitigation, geoengineering, what worked in the ozone layer, how did we solve that problem, and things like that.

But go for fission fusion. So I think fission is definitely, again, this is my sense, fission is definitely an important part of the solution. And I think it really needs to step up.

And I think apart from Germany, all the countries are going down that route. I think the generation four reactors, which are the small modular reactors and with thorium coming in. In fact, I think we are investors in nuclear fission and fission both itself.

So I think overall I am for fission specifically and SMRs. And this is like NTPC of India has also mentioned that they would want to look at those solutions. And fission, we have been doing it for the past 50, 60 years.

And if you look at the track record, it's pretty good in terms of... I think people are extrapolating something like one disaster with the repercussions over hundreds of years and counting that as the true cost. And that is mitigating the fission fusion effort.

I have a counter view to this, but go ahead. I know we've added this in the past. But I think again, these are the kind of narratives that are from my angle being created.

And that's why you need to look at the data. Because if you're not putting a power plant, nuclear power plant, you are effectively right now for the base load consuming fossil fuels. So in a way, by saying that this is not right, you are effectively allowing X number of deaths from air pollution on all those things to happen.

And in fact, I don't know the numbers, but it's like billions of tons of CO2 have been saved because there has been a nuclear fission industry. Coming to fission, fission has the classic joke of it's always 30 years away. But for the first time, it isn't 30 years away.

You think? So again, I don't think till 2050, it can create a scalable impact. Superficially, we are thinking fission might be possible if not in the next 20, 30 years.

I would say fission is not the solution to the climate crisis right now, but it's an important tool when you go beyond 2050 to 2100 for that society. But fission is a tool right now. Fission is, again, we can counter, but I think we should debate definitely.

But I think it is a tool for sure. You know, like he said, fission is always 30 years away. In India, there's also been a God-given number.

But for the first time, we had fission. But that's a net energy gain, not the net system gain. You know, the carbon budget for 1.5 is about 500 gigatons. Explain, what is 1.5? So 1.5 degrees. So to limit warming above pre-industrial levels is to 1.5 degrees. To 1.5 degrees? That is to limit warming to no more than 1.5 degrees. We have to limit the future combustion of fossil fuels to 500 gigatons.

Right now, to put that in context, we burn about 55 gigatons a year. So it's under 10 years of consumption, right? And there are variations.

For 2 degrees, it's closer to a thousand and so on and so forth. I don't think anything you can do on fusion will help you in this decade at the level of scale. Okay, that's the first thing.

Fission, there's been a God-given number in India on every energy planning document since the 1970s. We're going to build 50 gigawatts of fission, nuclear fission. What's the number today?

It just crept up from 6 to 7.5. The gestation time for a fission reactor is on the order of 10, 12 years. In India, it's often longer, right? Why do we think we can suddenly turn this around given that we haven't for all these years?

Plus the risks of accidents, plus the cost is low of renewable energy. And you have this scope for all these complementarities as we're talking about because it's modular. It's the same cost for a small unit, it's the same cost for 100x that unit.

Can't one argue that fission might be more scalable than solar? No, it won't be more scalable. There's an unmet promise of renewable energy in terms of development, right?

The idea that you can actually provide low-cost productivity services, as I was talking about, coal chains and so on, you can't do that with fission. You're flipping back to the big centralized power system structure with fission. And I don't think you're going to get anywhere near the scale that you need in the next 10, 20 years with fission.

So just one final thing, right? You know what our R&D budget in India is? It's about...

Of GDP, half a percent. So energy R&D is 45 billion, right? 150 million of that goes into clean energy R&D.

A third of it goes into fossil fuels. The rest is nuclear. Over all these years, 3% is going into renewable energy now.

That is less than... That's about 0.3% of what China spends and about 0.9% of what the US spends. We are backing completely the wrong horse.

I mean, the renewable energy story, the technology cost curve has been so steep. 80% cost reductions. We're cost competitive with coal.

And we choose to spend 3% of our energy R&D budget on this? But I think we're digressing into a different space. We don't have to innovate to figure out how to do fission better.

[Speaker 1] (1:58:37 - 1:59:17)

Somebody else can do it. But can't we replicate what someone else did across all of these technologies? Can I comment on the fission thing?

Very simple. To me, given the scale of the crisis, I think we have to think much more technology agnostically, in my view, okay? So I'm beginning to...

I mean, I'm not such a sort of only renewable, only renewable. Anything that will work, please bring it on. I'm just simply making the point from my point of view and not taking your position.

I'm just simply making the point that I would argue that if nuclear and fission can be part of it. So I'm just simply saying...

[Speaker 4] (1:59:17 - 1:59:18)

I just think it's the wrong horse.

[Speaker 1] (1:59:18 - 2:01:43)

I just don't think it's happening at the scale. And in spite of all the plans that I thought were exciting when I looked at China, looked at other countries, and looked at all the new modular designs coming up, they're not going anywhere. And to me, it is about getting either the PR right, getting the problems of waste right.

And on a flip, if I may, you know, it's a sort of... I find all the men get very worked up about fission and all the rest of it. It's a very masculine issue.

My attitude, therefore, is... Do whatever you want to do. Nothing's going to happen.

But do it if you want to invest all your political capital. But I think, just think of the potential of a technology that I think is happening under our noses, which can be transformative. So a city like Indore is taking its waste.

Now you're basically taking the waste that comes out of your households. Hydrogen. You are...

No, no. So you're taking the waste coming out of your... All this.

All this food waste that we will now waste. You will segregate it. We're going to eat it.

Okay, good. You will segregate it. You need to segregate it.

It needs to go in a segregated waste way to a collection and a treatment center. Now the problem with it always was there was no financial incentive for really making it pay. So now we are getting...

We've got this... I mean, there is technology which is old, which is not new. It's not rocket science.

It's a biomethanation technology. But essentially taking this waste and making gas out of it. Some compressed natural gas out of it.

Okay. Now it's... Just think of the problems that you solve.

You solve the problem of waste in your cities. Okay. You generate enough energy to run all your buses because in this case you can't really run cars.

So you also change the system because you say, if I have CBG, I have to have one or two places where it's going to be dispensed which means that I put them for buses. So just think of the opportunity. But let me...

[Speaker 2] (2:01:43 - 2:03:06)

I want to answer the nuclear fission. Couple of points. Firstly, this argument that it has not happened in the past, so it won't happen in the future is sort of a wrong argument because there are inflection points in anything as well.

And again, firstly, nuclear has definitely under-delivered. Like in the 1950s in the US, they were saying that you would not need to meter electricity. So that has never happened.

The dreams were... Hype was much bigger. That's for sure.

Do you think it will happen in 10 to 12 years, Merik? Okay. So the point is that my hope is around this generation four reactors which, again, there is a hope element to that.

But they should be getting online by the end of this decade. And they are more smaller than the big power plants. So the idea, the shift that is happening, the 10-year construction time frame, which is all because it's customized, you require a couple of gigawatt reactors or whatever, that whole thing.

What a lot of the new companies, including the ones that Bill is supporting, want to go for factory-made. So suddenly you go for 100 megawatt to 200 megawatt. I think the thought is, it's a different nuclear than what we are assuming in the past.

That's the only major difference. And for me it is, can that be plugged into a standard coal asset so it becomes a brownfield plant, not a greenfield plant, so that cost curve even goes down.

What's the raw material they're looking at?

The fuel? So currently there are...

[Speaker 1] (2:03:06 - 2:03:07)

Uranium and...

[Speaker 2] (2:03:07 - 2:06:20)

So uranium and... But thorium also. Thorium also people are looking at it.

So Merik, I think it makes sense. And I completely agree that we have to be basically trying... It has to be a multi-stranded solution.

There's no silver bullet here, right? I completely... So if nuclear becomes part of the mix, but given the scale of the transition here, I don't think it can be an either-or, right?

We have to do both these things. Exactly. And therefore, we are in a slightly difficult situation because you not just have to push, and I keep coming back to this, you not just have to push technologies, you have to put technologies, institutions, and politics that match.

What happens when that triangle looks different for renewable and looks different for nuclear? Then it becomes more challenging. And currently in India, they're doing both, right?

They're doing both. But I think we don't have a place, and this is a pet peeve of mine, we don't have a place in this government, and most countries don't, where you're actually thinking strategically about these pathways and what you need to put in place. This DISCOM story should have been identified a long time ago as a key building block.

It hasn't. So we're not thinking in that way. And that's why I get slightly nervous about the technology-driven solutions because they end up sounding like silver bullets, and the complications that have to happen to adopt them.

I'll just counter that. So just one final point I want to make. Something we haven't talked about here very much at all, and I've been trying to sort of bring it back a little bit to adaptation.

I think this is true in both adaptation and mitigation. The most effective way forward is shifts on the demand side, which India is really well-placed to take advantage of, because I think you were saying it earlier, Sunita, we haven't locked into things. We haven't fully built out our cities.

We haven't established our behavioral patterns. You know, how we build our cities, how we light, how we create architectural norms. These are things that are actually the cheapest with the shortest turnaround time.

So one of the interesting things is consumer technologies typically have much shorter technology turnaround times because they have shorter life than supply technologies. So you can actually iterate much faster. And for a developing country, those shifts in consumption patterns have to be about shifts that also enhance development, the kinds of things you're talking about when it comes to waste.

I'm going to bring it back to our target crowd, a 20-year-old boy or girl who wants to start a career here or a business here. In all the energy transition pathways we spoke about, what would you recommend? I think for my sense, bioenergy and everything around to do with bioenergy, waste management and circularity.

I think if you look at these three elements, which are sort of interconnected in a way because it promotes circularity, that's in India a very right problem to solve. And I think given the kind of audience also in terms of knowledge, better knowledge of the supply chain, better local on-ground implementation, operational challenges, that will be mine, at least.

[Speaker 1] (2:06:21 - 2:07:52)

So for me, my only advice is in India, I think what has really been important has been when we have been very frugal in our engineering, looking for answers that work for us. I mean, what Mirek said is what I would like to talk about, which is really, not just the fact that you take waste and you make energy out of, but you have so much happening in India where you're beginning to do things completely differently from the rest of the world. And I think that's the power of imagination.

I can name, I can tell you of many cases where people have taken on ventures, where they are doing upcycling of waste material, where they are doing, I mean, there is this young person I met recently who's doing this amazing job of, he's basically sourcing all the flowers that are thrown out of temples. And he's making agarbattis out of it, okay? There are huge opportunities, but again, those agarbattis are affordable.

They can be used by large numbers of people. They work for environment. I see incredible work happening in this country by the young.

And I think they're very different from my generation because they're much more connected to the question of climate change, to the need to do something urgently, but also much more nimble in thinking.

[Speaker 2] (2:07:53 - 2:10:29)

So I'll give you an area of opportunity. So I, and this comes back to something I said earlier. I think the scope for decentralized energy services that enhance productivity in rural areas.

Like the solar. Like, it's a really huge issue, but the general point I would make- You would suggest to a young person- Create a company. So there's this guy, we both- Which goes to farmers, takes a solar thing and says- So Harish Hande from Selco.

Selco is a great example of a small company where Harish Hande has talked about using renewable energy to provide development gains, whether it's coal storage for medicines and vaccines, whether it's for productivity enhancement, whether it's for agricultural processing. And figuring out not just, and this is where it comes up with the, it's not that I'm technology phobic by any means, right? But you have to build out the institutional structure, the case, you have to make sure that the politics of it work, that the technology isn't going to be strangled because a more powerful interest is going to cut it off, right?

You have to make all of that work. I think that renewable energy, because it can be targeted, can be used to increase the purchasing power and the livelihood opportunities of rural consumers in India. And I think there are lots of ways of, I mean, it's not what I work on directly.

So somebody like Harish would be a much better sort of person to talk to about this. But I know that that's important. But the general point I would make is- But that's a good idea.

Like if I'm a young kid, I can figure out how to package a solar, go to a farmer, share the benefits, maybe sell it as a subscription. Is that financially viable? I don't know.

Again, somebody like Harish has run businesses in this way, so he would know. I suspect the answer is that it depends. But it's a twist on this idea of energy service companies, right?

Which have been a really powerful model worldwide where you're not providing energy, you're providing energy services. There's another interesting company, what they are trying to do is charge the bore wells. Exactly.

It is like similar, like take the services and provide a service to the- The bigger idea, though, is you've got to find ways of bringing a development problem in conversation with the climate angle, right? It's got to make sense in terms of the development choices we have to make in India. Whether that's on the mitigation side or on the adaptation side, something that makes you more resilient.

So that's that sweet spot, that intersection between these, that's where I think we have to look. Bhumi, you want to weigh in on this?

[Speaker 1] (2:10:29 - 2:11:43)

I'll tell you what I do, right? Because I've just kind of started my journey where I independently try and support as many green businesses as I can, right? It could be what you're using over there, which is like- Green tissue paper.

Green tissue paper or something that I read recently where cow dung was used as building material, right? So I'm constantly on the lookout for companies that I can support. When you think of what Young India is doing, right?

You constantly see disruptive innovation. There's somebody who's literally making ink out of the carbon that's available in the air. There's Thali that's actually recycling all the single-use plastic available.

So I feel like just looking for solutions that are around us for problems and actually taking, as you said, what they do with waste management. I think that's such a great space to just- Even if I was somebody who, as I said earlier, I've started onto my investor journey, I'm constantly looking for opportunities that kind of help give a solution in a way where we don't create more of the wastage, but it consumes it all in there.

[Speaker 2] (2:11:43 - 2:12:04)

Interesting. Another question, if I may ask you guys, if I'm a young person and I want to participate in conversations around this, like I want to talk to you on a forum, I want to talk to you on a forum, I want to talk to you guys, where do I go? You should read down to earth.

And COVID has actually really expanded ironically.

[Speaker 1] (2:12:05 - 2:12:05)

Online forums.

[Speaker 2] (2:12:06 - 2:12:12)

Everybody does webinars. Which ones? Everybody does webinars now.

Like two-way conversation, if I want to be a part of the community.

[Speaker 1] (2:12:13 - 2:12:17)

I think a lot of our institutions are doing open webinars.

[Speaker 2] (2:12:17 - 2:12:35)

Can you name some good ones, like the funds that come to your mind first? Well, I mean, I think all of our organizations, where I was the Center for Policy Research, the new organization we set up, Sustainable Futures Collaborative, but TERI, all the others in our ecosystem, everything is now open access in terms of webinars.

[Speaker 1] (2:12:35 - 2:12:37)

We reach out to huge numbers of people.

[Speaker 2] (2:12:38 - 2:12:46)

I'll take a different lens, like a lot of climate tech VCs are doing mixers. Specifically, I'm talking about Bangalore and Mumbai areas.

[Speaker 1] (2:12:46 - 2:12:46)

What's a mixer?

[Speaker 2] (2:12:47 - 2:12:52)

So, it's basically a sort of a dinner, not dinner, but it's people meet up.

[Speaker 4] (2:12:52 - 2:12:53)

How they have singles mixers.

[Speaker 2] (2:12:53 - 2:12:54)

Yeah, sort of.

[Speaker 4] (2:12:55 - 2:12:56)

Climate dating also is something.

[Speaker 2] (2:12:57 - 2:13:07)

I mean, is that happening? I think you could have a singles climate mixer. I think it's happening.

A sort of speed date, that's also come up. Really? It's happening, I think.

[Speaker 1] (2:13:07 - 2:13:09)

There is a lot of... Really, this world is not something.

[Speaker 2] (2:13:09 - 2:13:55)

How does one find it? So, there are like few, if you say, if you follow the venture capitalist in the climate space, I think a lot of them do this. Can you name some?

I think Avana, for example, I know definitely does some of this. Avana, I think, is one of the largest in the space. Then Special, which is into deep tech, they would have some.

Then Transition VC. I think everyone would have some sort of a mixer sort of a thing. I think there is...

Then Sustainability Mafia is there. They do some events. There are some events which happen and maybe then you can go in personally where the two-way communication starts happening.

And then you meet someone there. Interesting, okay. Next topic.

What did we do right with the ozone issue? Ozone was like this world ending issue many years ago.

[Speaker 1] (2:13:55 - 2:14:08)

It's very clear. I mean, it's a very simple story, but a very clear story. It does not have the lessons for climate change.

So, that is something we need to start and understand very clearly.

[Speaker 2] (2:14:08 - 2:14:10)

May even have set us on the wrong course.

[Speaker 1] (2:14:11 - 2:15:47)

Oh, really? What has happened with ozone actually, not only does not work for climate change, but we may have exactly as Navroosh is saying, we may have derailed the climate change action because of the action. So, what happened in ozone?

We had a chemical made by a multinational which was destroying the world's ozone layer. That same multinational came up with an alternative which only made sure that the ozone layer was less destroyed, okay. But there was an alternative technology available from the same multinational, okay.

And many more such companies that came on, DuPont and many others that came up. Now, the global consensus around it was we need to do something because the ozone layer is being damaged. We need to repair it.

For that, we need to phase out this technology. And to phase out this technology, we need to phase in the new technology which now is being held also on proprietary grounds by this same company which made the first technology. So, what was the deal?

The deal was that money would be put in to buy the technology which was in proprietary hands. Public money would be used and that technology would then be given to countries which had a 10-year lead time to catch up. So, you took public money.

You had a problem that was created by a company. That company was not held to account.

[Speaker 2] (2:15:47 - 2:15:47)

Which company?

[Speaker 1] (2:15:48 - 2:16:19)

DuPont, okay. They were not held to... ICICI and DuPont.

They were not held to account. They were not told that they had damaged the ozone layer. Polluter pay should operate.

They should be asked to pay for it. In fact, the global agreement was that now that you have found the new technology, we will buy that from you and replace your old technology with it. Okay?

And we have gone through four variations of this now by the time we have reached today. HFC-134a, which now HF4s are coming.

[Speaker 2] (2:16:19 - 2:16:20)

What is HF4?

[Speaker 1] (2:16:20 - 2:16:22)

Each one of them are new chemicals.

[Speaker 2] (2:16:22 - 2:16:26)

So, you start with chlorofluorocarbons, the original ones.

[Speaker 1] (2:16:26 - 2:17:34)

Then you add CFCs, then HFCs, then hydrofluorocarbons. So, each one has been a substitute to the other. Companies have made profit, laughing themselves to the bank.

There's also a twist. The public sector has paid for those technologies. Countries like us have been told that if we...

And the operative mechanism was that either you join, we give you 10 years to join, and if you don't join, then we will put trade measures on you. So, essentially, there was a deterrence to say, let's put a trade measure. We won't trade with you.

So, now you make your air conditioners using the old technology. We won't buy that from you. So, it was a clean deal.

Everybody went laughing to the bank. Yes, the ozone layer has repaired itself to a large extent, but the technology, the chemical which has come after all these variations is now being understood to have major impact on climate change. So, we started off at CFCs.

We're at HF4.

[Speaker 2] (2:17:34 - 2:17:35)

GWP potential is very high.

[Speaker 1] (2:17:35 - 2:17:36)

Is very high. Now...

[Speaker 2] (2:17:36 - 2:17:39)

What is GWP? Global warming potential. Global warming potential.

[Speaker 1] (2:17:39 - 2:17:46)

So, very bad for climate change. These guys love technical work. It's bad for climate change, okay?

The problem I have...

[Speaker 2] (2:17:46 - 2:17:47)

Worse than any of the other gases.

[Speaker 1] (2:17:47 - 2:18:57)

The problem I have is that this ozone story became... Because it was a Western-built, companies loved it. It's sort of, you know, everybody has gone laughing to the bank kind of story.

It started getting applied to climate change. So, we were told we'll get 10 years more to do the reductions. Obviously, not possible at the differences of the emissions from there and the emissions.

We needed to increase. But more than that, what we have done is to basically not work the reverse. I mean, to me, the ozone story was one where the rich producers, the rich consumers used a stick on the poor consumers, producers, and said, if you don't do this, we will beat you on the head.

The point that I have is, on climate change, where is the reverse, okay? We can't use trade. We can't use any of the levers to basically demand action for them to reduce their emissions.

So, ozone is not a story that needs to be understood from a climate lens.

[Speaker 2] (2:18:57 - 2:19:09)

I mean, just to add to that, at a fundamental level, this is about essentially a niche chemical made by a very small number of industries. I didn't realize it was made by one company.

[Speaker 1] (2:19:09 - 2:19:15)

It was just a few. At that time, two. DuPont and ICICI.

Now more have come up. Indians have...

[Speaker 2] (2:19:15 - 2:19:46)

Under-licensed to others and so on. So, the difference here is, it's not just carbon dioxide, but substantially carbon dioxide. Carbon dioxide and fossil fuel burning is the core of the industrial revolution and human development.

Human development has co-evolved with fossil fuels. It's baked into everything we do, all around us, all the time. Cement, lights, transport, wherever else it is.

So, the nature and the scope of the change is just so much, so much larger.

[Speaker 3] (2:19:46 - 2:19:47)

It is at another level itself.

[Speaker 2] (2:19:48 - 2:20:19)

That in fact, and I think this is where the shift in mindset is important with climate change. Yes, it is a global collective action problem in the sense that, you know, we have to, you can free ride and you can choose not to act and wait for somebody else to act and so on and so forth. But I think the way we are thinking about it more and more is, can you shift your economies in ways that make sense for your economy and have the effect of solving climate change?

Can I, can I lead this? That shift, that was not possible with ozone.

[Speaker 1] (2:20:19 - 2:22:03)

One interesting fact, which I think is a little different from ozone, climate and all, is about, so in Hindu religion, it's very interesting, the God of destruction, Shiva, is more important than the God of creation, okay? Now, if you look at it, which are the products that are creating problem in our world? So we started off with the biggest environmental problem that we've had much before climate change was DDT.

DDT was a pesticide. Why was it a problem? Because it did not destruct.

In fact, when it was discovered, DDT, the person got a Nobel Prize for it because it was a

pesticide, a chemical that could not be, was not destructible, okay? Then came CFC. Again, got a Nobel Prize.

Why? Because it could not be destroyed. It was stable, okay?

Plastic today, can't be destroyed. So actually our biggest problem is coming from the fact that we as human beings are still rewarding something that nature actually tells us should not be rewarded. Nature says that it should be destroyed.

It should go back to earth. And yet we're the arrogance of man, because I don't think any of the women were involved in any of these products that were designed, okay? Was to make products which are not destructible.

And I think that's really the fundamental problem that we have today with environment.

[Speaker 2] (2:22:03 - 2:23:38)

Can I take this for carbon capture? Or CO2. Carbon capture projects.

Do they have a future? Are they legitimately a possibility that can curtail some of the ill effects? So like in carbon capture at source, definitely has some role to play.

Because again, in cement you will require when you go from CO2 to chemicals and say sustainable aviation fuel, e-fuels and all those, that is definitely there. Direct air capture, which is there, I think it is just too energy intensive. I think it is like the current numbers are it's three megawatt hours.

Again, not taking the numbers, just too energy intensive. Right now, I really don't see a pathway to energy intensity reducing so that it can be really scaled up until some new technology comes in. I'll give you a funny story around this.

There are people in Bombay, some kid pitched us this company which is doing carbon capture and he's putting it in parks and stuff like that. People in places like Delhi, Bombay are attempting it, young guys, which is quite interesting. No, you can.

So basically, fundamentally, it's not a difficult technology. If I take a, again, slightly technical, if I take a base and just rotate around it, you will capture some carbon dioxide because it will react to it and you get it captured. So it's not an impossible technology, it's a simple technology.

But is it scalable? That's when the energy budget comes in and then it's quite high so far. Having said that, in a lot of the IPCC reports, I don't know, but they have mentioned that you require some sort of carbon capture as well in the later years.

[Speaker 1] (2:23:38 - 2:23:54)

So, planting trees captures carbon as well. I mean, this is the point. There's nothing against carbon capture.

The point is, what method are you going to use? How energy intensive is it going to be? How affordable is it going to be?

How scalable is it going to be?

[Speaker 2] (2:23:54 - 2:24:02)

I think the key element here is, you're just doing carbon capture using fossil fuels, not like trees which are using solar energy.

[Speaker 1] (2:24:02 - 2:25:26)

So carbon capture using fossil fuel is a very different thing, where you're saying, I have a power plant, I have a power plant and I basically capture the carbon dioxide. And then, I mean, what's interesting now is earlier on it was storage and those storage projects haven't gone anywhere. There's been a lot of talk about them.

Norway is doing a lot of deep sea storage. There's a lot of talk about that. Australia has a lot of projects, some projects that are, but nothing has come to fruition.

Nothing has come to scale. But there is a hope that those projects will happen. But there is also now new, more and more thinking, and I think more innovation, that how can we utilize that carbon?

So instead of storing it, how can we utilize it? Now, I mean, to me, we need to be open minded at the possibilities. I mean, who would have thought that shale gas would happen at the scale that it has happened and taken?

I mean, that the Americans today are the world's largest producer of oil and gas. They've beaten Saudi Arabia, okay? So, I mean, that's something that I had not thought about with all the oil shocks and everything.

So technology has a role to play. But I think from all that you've heard from all of us, it is about affordability.

[Speaker 2] (2:25:27 - 2:26:47)

It's about scale and speed. So I, you know, yes, but with an important, I think it's important to understand where this comes from, right? It's not the interest in carbon dioxide removal and also I should add solar radiation management, which is the far more risky geoengineering side of it.

In a sense, it doesn't come because there's been some big technological breakthrough. It comes

from two things. One, when the IPCC runs these models and we say politically, we want to try and keep to a world of 1.5 warming, the models don't close. You can't do it without a whole bunch of assumptions around carbon dioxide removal. So the modeling and the knowledge is driving this. That's the first thing.

The second thing that's driving it is the politics. So in all the negotiations and in the IPCC, the oil producing countries, every mention of low carbon development or mitigation, they include or remove them because that's a way for them to keep pumping oil, right? So the politics has opened the door to that, which means that now the technology development has, it's sort of amplified, right?

So these are two of the reasons why we spend so much time and energy. It's not that the technology is mature, really.

[Speaker 1] (2:26:47 - 2:27:20)

But on the other hand, Navroz, I mean, given the fact that, I mean, I really struggled with this. The whole question about abatement technologies, okay, abated coal, abated. So this is the new politics in our world now, because the word is now not fossil fuel phase out.

It's about an unabated fossil fuel phase out, which means what is that fossil fuel where you have not done carbon capture and storage or you have not been able to clean it up.

[Speaker 3] (2:27:21 - 2:27:22)

At the time of burning.

[Speaker 1] (2:27:22 - 2:29:32)

At the time of burning. Now, to me, and I argued with this with a lot of people, including my colleagues, who are my toughest sometimes on these issues is, you know, we need to, if we say just fossil fuel phase out, it's not going to happen. In fact, it's not happening anywhere in the world.

Forget our world. I mean, our world is different. It's not happening in the world it needs to happen.

They're going back to fossil fuel more and more. Their drill, baby drill is, you know, Trump may talk about it, but it's Biden who's opened up the federal lands for drill, baby drill. OK, so shale gas export.

I mean, America is today the world's largest producer. I mean, it just galls me that they preach to us that fossil fuel phase out because they are talking about coal and not oil and gas. I mean, it's all fossil fuel.

So in that language, I think it's important for us and that politics. I think it's important for us to

challenge the oil and gas and the thermal power industry to say, you know, OK, so you do have a future in the next, but you have to do a bit. OK, now it becomes, I mean, I've used this also in India because so we did a report some years ago looking, we rated thermal power plants in India and basically found they were terrible in terms of emissions.

I'm talking about local emissions. I'm not talking about climate change. I'm talking about particulates, NOx, SOx and, you know, CSE tends to name and shame.

So we had named all of them as well. Who was bad? Who was better?

What was it? It had huge impact. I mean, governments immediately reacted.

We got a committee. We got standards set. Industry was told to meet the new standards.

Now, a lot of my colleagues said, but, you know, you are giving coal plants a new life by asking them to clean up. You should be asking them to phase out. And my point was they will not phase out, but we need to clean them up.

So, yes, it is an oxymoron to say clean coal. But on the other hand, if we don't clean up coal and we don't do it fast, we are going to be in trouble.

[Speaker 2] (2:29:32 - 2:30:14)

I would agree with you. And this is a very interesting point because there are some interesting technologies there, say you use supercritical water in the plant to just increase the efficiency. Now, this is again that whole point that if I increase the efficiency of a coal plant, it's actually in a way helping in the near term for sure.

But maybe you are also extending the life. And these are the kind of interesting, complex challenges which will have to be faced. And I think with unabated, it's I'm not sure I'm fully on board with you on this, Sunita.

I might be with your colleagues, because the reason is that all the abated coal plants are going to be new plants. So you're basically, and they're basically by and large, right? Because many of the old ones are not ready.

[Speaker 1] (2:30:14 - 2:30:16)

So retrofitting is possible also.

[Speaker 2] (2:30:16 - 2:30:22)

Yeah, but it's the economics, I think, is more for you. Maybe you know more about this.

[Speaker 1] (2:30:22 - 2:30:26)

So the gas, actually, it's even harder to abate gas than coal.

[Speaker 2] (2:30:26 - 2:30:35)

But just to say that the risk is, and also abatement, but I understand the definition is 80% abated. So you still have 20%. And you're extending the life.

[Speaker 1] (2:30:35 - 2:31:03)

Anyway, this is getting into the final draft. The larger question is, I think, let's not get into the technical issue. The larger question is, they're going to be, we're going to have to have very real solutions.

They need to be scaled up. Many of these other answers will need to work in. But I think from what Navroz is also saying, none of those solutions can be half-baked anymore.

That you can't use abated as an excuse to continue polluting. You need to then...

[Speaker 2] (2:31:03 - 2:31:53)

Okay, I'm going to get into the last two things for tonight. Geoengineering first. It could be adding sulfur in the stratosphere, covering ice with limestone, trying to tweak the after effects of climate.

Thoughts? Anybody? I've never looked at it.

I have no clue. I think there is sulfur, stratospheric sulfur, aerosol injection. Then you have marine cloud brightening, which is also an interesting thing.

Ocean seeding. Ocean fertilization. With what we are burning with fossil fuels, we are effectively running a large geoengineering experiment itself, right?

So that's also an experiment. My thought process is, we should do all the research. I am not against doing all the research.

I think some people have gone beyond research. I would right now at least draw the line.

[Speaker 1] (2:31:53 - 2:32:00)

What is the arrogance of man? I mean, to think that you could tinker around with nature to that extent, that you could fix...

[Speaker 2] (2:32:00 - 2:32:01)

But they tinkered...

[Speaker 1] (2:32:01 - 2:32:15)

But to begin with... To fix what you have so badly broken. And you have completely disrupted nature today.

You've gone way beyond nature's absorptive capacities. It's the arrogance of man which has brought us to this stage.

[Speaker 2] (2:32:15 - 2:33:16)

But we're not saying we should not research it in time. And we think that the arrogance of man is going to get us out of it. I think I agree with you.

So my sense is, what are the risks? Like again, the bigger dystopian future that you ask the question. Like there are certain tipping points of climate which you may hit if you are not doing...

if you are doing business as usual. Now, whether it's the permafrost thaw or the western Antarctic ice sheet and all that. So I personally feel you should do the research for sure.

It's worth explaining a couple of these tipping points, I think, for the audience. It's quite interesting. Go for it.

Permafrost. So basically, permafrost is frozen soil, largely in Siberia and all those regions. It has more carbon stored in the Siberian permafrost than the whole of atmosphere right now, which is a pretty big number.

That's one of the tipping points. Then other can be the west Antarctic ice sheet, which is right now, there's a Thwaites Glacier there, which if it goes, then the whole west Antarctic ice sheet goes in the water and all that. Which also means that the reflectivity of the earth goes down.

[Speaker 5] (2:33:16 - 2:33:17)

So more gets absorbed by the water.

[Speaker 2] (2:33:17 - 2:34:00)

And there is a risk of cascading tipping points again. If you start hitting those, then we are in a different league of problems than this heat strokes and the droughts and all this we are talking about. That's why my sense is do the research.

But the deployment and the governance has to be very, very, very strong. Again, there may be struggles in the past. Yeah.

So in my world, there's a lot more conversation about the global governance of geoengineering. That essentially, what you don't want is countries going cowboy and doing their own experiments. And that it's and how do you actually incentivize countries to stay within the fold?

I think we are at the stage that we need to know about it. I think we're getting to that stage where we need all of it.

[Speaker 1] (2:34:00 - 2:34:37)

No, I don't think so. I think there are some things that define. I mean, I'm not against new technologies.

So I'm not a Luddite. I'm not sort of saying that, you know, all this new technology, it's all great. But there are some technologies which basically have the same principles which we are fighting against.

We are fighting against anything which disrupts nature. You can't then find an answer to nature's problem by disrupting it even more. So to me, there are certain things which are my thresholds.

[Speaker 2] (2:34:40 - 2:35:12)

You alleviated to this earlier when you spoke about buying carbon offsets. What do we think about the idea of carbon tax? Putting a price on carbon from straight economic logic is a good thing.

And a carbon tax is a better way to do it than a cap and trade system, right? No question. In my view.

But how do you tax it? Do you tax it on carbon consumption? You can just tax the fossil fuels itself.

So then that's the simple answer. You would tax it at the play, at the choke points. Basically, the fewest points, which is at production, basically production, import, et cetera.

[Speaker 1] (2:35:13 - 2:36:45)

No, but the CBAM is that. So that's Europe's answer. Europe has basically put carbon adjusted border mechanism in place.

It's too late at night for me to get CBAM right in terms of it. Essentially, it's a measure benign. If you look at it to say, oh, we are putting a tax to make sure there's a level playing field so that the Indians who don't have clean technology cannot come into our markets.

But it's actually a trade measure being put in by the Europeans to stop the import of a lot of products, which they now want to grow their own industry in. And it does put the whole, I mean, the point is globalization hasn't worked, as you said, that elephant. But the problem is that we are all wedded to globalization today because we are export oriented industries.

We don't have any other economic system. So in my view, these kind of taxation systems that are

emerging in the world, the U.S. has IRA, which is a subsidy system, works the same, because you're subsidizing your own industry versus other industry. Europe has a taxation system.

I think those are going to be bad for climate change. We need cooperation in climate change, not competition, not polarization. We need the world to come together.

[Speaker 2] (2:36:45 - 2:37:20)

And to that point, I think one of the sense that I have is the world, every country or region will have its own strength. And if you're not, because the problem is so big, if everyone is saying, I want to do everything myself, I want to do manufacturing myself, I want to do research myself, everything at myself, I think that just creates inefficiencies at a time where we don't have that. We really need to have that cooperation across the board.

The catch-22, though, I agree with that, is that each country has to figure out a way to make this sell in its own country. And the way to do it is to try and keep the benefits at home. That's what we're finding.

Bhoomi, would you like to weigh in on this? Carbon tax, what do you think?

[Speaker 1] (2:37:20 - 2:39:33)

I mean, we actually kind of spoke about it before we started the session, right? It was something that I asked them. I was like, shouldn't somebody take ownership of it, right?

Because there are these large corporations, there are the developed nations, there are the large polluters, the richer spaces, there are the large polluters. So shouldn't they take some ownership and, for the lack of words, give back? So this was actually the exact question that I had in my mind, which now I kind of get some clarity from her.

But Bhoomi, the point is that that's the right approach. I mean, they are polluters. They need to pay for their pollution.

Yeah. Okay. But the fact is that as they pay for the pollution, it will raise the cost of their production.

In my view, there's nothing wrong with it because that then incorporates the price of environment. It will also reduce consumption and it will probably make the world better off by doing all those steps. Okay.

The problem is in an unequal world. I mean, what I love today is Indonesia is incredible. Indonesia has discovered nickel.

Okay. And it's basically said we will not export nickel anymore. We are going to develop our own industry, our own battery industry.

You want battery industry, you come here to invest. Okay. Now, actually, the Indonesian case has been taken to WTO by the US and other countries.

How do you have the right not to export your raw material? Now, if you could actually build a counter to globalization, a real counter to globalization by making it a really effective localization movement, okay, then you could actually see a difference. But today what's happening is you have a very unequal world where the rich today have decided the rules of globalization don't work for them and therefore they want to change the rules.

But they're changing the rules only so that they work for them and not work for the world, not work for the planet. And I think that's the problem with carbon tax. Good idea, bad in implementation.

[Speaker 2] (2:39:33 - 2:39:48)

I think we are in this, we haven't talked about this yet, but we are in this world where the global regime is built around the so-called nationally determined contributions, right? Because we couldn't get a kind of a division at a global level.

[Speaker 1] (2:39:48 - 2:39:49)

Because we didn't fight enough for it.

[Speaker 2] (2:39:50 - 2:39:50)

Okay.

[Speaker 1] (2:39:50 - 2:39:51)

I mean, I'm sorry.

[Speaker 2] (2:39:52 - 2:39:55)

Can I explain the NDC story?

[Speaker 1] (2:39:55 - 2:39:59)

He and I completely disagree on this. This is one issue on which we never will agree.

[Speaker 2] (2:40:00 - 2:40:47)

What is the issue? Well, I'll explain the concept and then maybe I'm sure Sunita will explain why she disagrees. We won't get into why.

The gist is that every country, every country bottom up agrees, every country states what it will

do. And right now we are in a world where a country can decide whether to use the carbon tax to meet its domestic pledge, right? The conversation about what one should do cooperatively, globally has dissipated in the kinds of solutions like Tobin tax, which in principle is a great idea.

I really think it would be extremely helpful, particularly around, say, money for loss and damage, right? To most impacted communities. That global deal has broken apart post-Paris.

[Speaker 1] (2:40:47 - 2:44:16)

So the issue of differences, I very strongly believe, and that's my, no reason for anyone else to believe it or accept it, but it's a very strong belief that the key reason why we are not acting on climate change is because we broke a global rule-based system. That we needed to have the rules created, a global rules created for countries to act. The principle that was done in Rio at the Framework Convention on Climate Change, the first agreement, global agreement, followed through at the next protocol, which was the agreement set up called the Kyoto Protocol, where you essentially agreed that here is a group of countries which have overemitted.

Here is a group of countries that needs to grow, development. So let's agree on a target for these countries to reduce. Let's agree on financial and technology transfers so that these countries can grow, but grow differently.

But there were rules set up. Weak rules, because the US and others basically weakened it. Compliance was weak.

Deterrence was weak. But still, there was a framing. Now, this was too inconvenient because, obviously, rule-based systems are tough for the world to accept, particularly large polluters.

So rules are very good for the poor and justice is very tough for the powerful. So that system was taken apart at the Paris Conference, which happened in 2015. You would have heard of the Paris Agreement.

The Paris Agreement essentially said, forget that. That's too complicated, not complicated. It hits into our sovereignty.

Countries won't follow it, which means US won't follow it. So let's agree on rules in which everybody will do what they can to save the planet. And everybody who will do what they can to save the planet is looking at a three degree centigrade rise over 1850 and worse.

And this, to me, is the problem. So when we started the Kyoto Protocol in the 90s, you didn't have China. China was part of this part of the world, our part of the world, because it didn't have emissions, it didn't have growth.

Now, if you had a rule-based system, you would have set rules to say, if a country crosses this threshold, it will have to take on these emission reductions. So you would essentially build a

system in which the, and it would apply to India as well, that any country which crosses the threshold, we don't have that today. So China will, in 2030, equal the emissions of the United States on a per capita basis, per capita.

It's already double, it's five, they're 10, okay? They will equal the per capita emissions. Now, this is why, in some senses, the climate debate really needs to go back to global governance.

You cannot deal with an issue as much which is about the globe without a system of global governance.

[Speaker 2] (2:44:16 - 2:44:22)

What is the consensus here? Is being vegan, vegetarian, good for the planet, bad for the planet?

[Speaker 1] (2:44:22 - 2:48:34)

So I put out, I have very clear views on this and I got a lot of flack from a lot of people in India because of what I said, but that's my view. Which is? So my view is, the way we grow food is the question.

The amount we eat is the question. What we eat is the question, okay? It's not about vegetarianism.

The fact that we are growing tomatoes in this crazy manner, in fact, our latest issue of Down to Earth has a cover story from where ICR scientists have done this amazing work looking at wheat and rice and shown how they have lost all their nutritive values over the years. So you're growing food in a way that you are manufacturing food. Now, that itself, you're chemicalizing it, you're putting poisons into the soil, you put antibiotics.

We tested, as you know, we have a lab, so we tested honey, we found antibiotics in it. We've tested chicken, we found antibiotics in it. So those are the problems, the way we grow it.

Then, how much we eat, okay? The amount of meat that is consumed today, not forget the fact that that meat is grown firstly by clearing rainforests using huge amount of water, all the rest of it, then the amount you eat, okay? And the amount of waste you have.

Now, that's not the question in India. The question in India, it's not about vegetarianism because the fact is a farmer in India that keeps a cow, buffalo is a better example because veganism is all about milk and all the rest of it. So that farmer can get a value from that animal, okay?

Now, in India, we follow a farming system in which that farmer keeps the animal, the dung of it is put on the land, you get fodder, which then comes from the neighboring. So it's a, in technical terms, it's called an agro-silvo-pastoral system. It's an agricultural system, a pastoral system, and a forest system.

In that, the animal is critical. For climate change, the animal is critical because the only risk mitigation strategy, the true adaptation strategy that a farmer has is to make sure that the risk is not just in the crop but also has an ability to keep animals, which then give it meat, which give it milk, which give it dung, okay? Now, if you believe in the kind of vegetarianism, veganism, which is becoming an issue, what happens to the Indian farmer?

You demonetize their cattle. You take away their ability to be able to use their animal, their livestock, for the value that they will get for it. I mean, I don't know how many of you keep cattle, but I can tell you from what I heard from farmers, it's not cheap.

And you keep them only for the milk-rearing period, which is very small. And then if you're a vegan, you don't even have the milk. So I think we need to get the food debate a little right and out of this sort of, you know, it's late at night, so I don't want to say something which will get me even worse pressed than I normally get.

But we need to get it out of all this sort of, you know, I'm a vegan, so I'm an environmentalist. No, you have to talk about how much you eat, how the food is grown, and what benefits does the farmer get? The problem is in the food debate, nobody wants to talk about the industrial farming systems, the intensive food farming systems, the agribusinesses that control the food of the world.

They control the way the food is grown, what is supplied to you, the kind of glamour that is built in to the food that we eat. I mean, we have a huge campaign against junk food.

[Speaker 4] (2:48:34 - 2:48:35)

Can I ask a question?

[Speaker 1] (2:48:35 - 2:49:04)

We get so much, I mean, I cannot tell you the power of those companies in supplying us the food that we eat and how, we haven't lost the battle yet, but we will probably, one of these days. But we were hoping to get a label on the food that we eat, the junk food that we eat, which is simple enough to tell you that when you drink one bottle of Coca-Cola, you have three days quota of your sugar.

[Speaker 3] (2:49:04 - 2:49:05)

Okay.

[Speaker 1] (2:49:05 - 2:49:37)

Or if you have one packet of Maggi noodles, you have a whole day's quota of your salt. So if a child eats a Maggi noodle, one packet, they must eat boiled vegetables after that for the day.

Okay.

That's the kind of, that's the kind of salt, sugar, fat that is going into this food. That's the nature of the agribusiness. Huge, powerful.

I mean, we talked about tobacco. This is the real tobacco of business.

[Speaker 2] (2:49:38 - 2:50:21)

It's a very interesting point. Can I just ask a question? I just, you know, trying to tie some of these things together.

It occurs to me that, you know, because climate change is in a sense, the first environmental issue that has become kind of geopolitically important, right? It's the subject of G8 meetings and G20 meetings and whatnot. I've often wondered to what extent, for those of us who care about environmental issues more generally, the whole food system and so on and so forth.

Can one use climate change as the way to actually open up some of these other conversations? And it's this idea of climate change is kind of really not a thing in and of itself as a connecting tissue. It's how do you open up urbanization conversations?

How do you open up food conversations?

[Speaker 3] (2:50:21 - 2:50:22)

No question.

[Speaker 2] (2:50:22 - 2:50:22)

Waste conversations.

[Speaker 1] (2:50:23 - 2:50:27)

But the question is not just open it. Not just open it. Follow it.

Take it. No, absolutely.

[Speaker 2] (2:50:28 - 2:50:41)

But, you know, we've often been stymied because there are very powerful interests. There's no way into these things, many of us. So climate change is a way of forcing it open and then trying to change things.

And there's a lot of talk about intersectionality that happens.

[Speaker 4] (2:50:41 - 2:50:43)

That's exactly right. That talk is getting more and more.

[Speaker 2] (2:50:44 - 2:50:52)

That's what I'm getting to. Climate with whatever. Education or climate with health.

But I think that's true. Financial reform. Climate, you need to solve all of them together.

That's right.

[Speaker 3] (2:50:52 - 2:50:53)

The food system.

[Speaker 2] (2:50:53 - 2:50:59)

But I think we might disagree on the solutions a bit. But prima facie, food system has to be transformed.

[Speaker 1] (2:50:59 - 2:51:38)

But I think to take forward both of what you're saying. Yes, we need it. And it's happening in the world.

We are beginning to see the interfaces. My problem is that you're going to get a narrative which is strong and powerful, which is coming from the other side, which is going to derail the real action, the need for change. OK, now that's where the strength has to be.

Your ability to stick out, say something which is inconvenient, be able to back it up with facts. Be able to argue for it and stick with it. We're all becoming, I know this is going to kill me because I love saying this.

We're all becoming gentlemen.

[Speaker 2] (2:51:41 - 2:51:48)

But in summary, this point, the takeaway is saying that you're vegetarian to save the planet doesn't hold much weight.

[Speaker 1] (2:51:48 - 2:52:03)

It may in the north, where you're eating a lot of bad food, too much meat, growing it badly. But if vegetarianism means that you eat vegetables which are also grown badly, it doesn't help. It's a bit more nuanced.

[Speaker 2] (2:52:03 - 2:53:02)

And context matters a lot. OK, so the last point of the night, I want each one of you to give me one, one, just one liner advice to the government, what they can change and one liner advice to a young entrepreneur or someone starting up. You can start, Miric.

To take Sunita's point, carbon labeling is something that the government should do. I think that will help consumers take decisions better. So that's one.

To entrepreneurs? I think to entrepreneurs, my sort of advice would be sometimes they go after the solutions in that this is a good business-making opportunity and all that. Just ground that on the problem.

Just don't look at the solution. And sometimes if you only look at the solution and the business side of it, you may lose fact of that you are trying to solve a problem in climate if you are starting your journey. So ground all your businesses with a problem of climate change and then go for the solution.

Bhumi, would you like to go next?

[Speaker 1] (2:53:03 - 2:53:55)

I think for me, it would be more about something that we spoke about again earlier, is that how do we get our populace to adapt to the change that's coming. That's something that I'm completely unaware of and I think most people are. That okay, if there are such massive changes that are happening on an educational level, I think it needs to be taught and spoken of.

And I think for anybody who's starting their entrepreneur journey, I think supporting green businesses like many of us do is something that could kind of help find solutions and just supporting innovation. Or starting green businesses. Starting green businesses.

Sunita? So, I mean...

[Speaker 2] (2:53:56 - 2:53:59)

One advice to the government that they can change? One to the entrepreneur.

[Speaker 1] (2:54:00 - 2:54:50)

To government, the act at scale on the plans that you have, implement them and to make sure that they actually make the difference that we can make. So that would be my... I mean, just get determined behind it, scale it up.

So that would be... And to an entrepreneur, see, I don't know... I mean, green business is important.

To me, every entrepreneur is important because anybody who's in any business has to be an

environmentalist. So if you're running an iron and steel industry, or you're running a cement industry, or a textile industry, or an industry which is a new industry, all of them have to incorporate sustainability in real terms and not just in words.

[Speaker 2] (2:54:51 - 2:57:27)

Navrooz? So to me, the advice to the government is to create bodies such as a low carbon development commission that systematically looks for these intersectional opportunities that we've been talking about, food and climate, buildings, urban, etc. And perhaps also wraps that up in a climate law that incentivizes this linkage between climate and development.

I think this sort of intersectionality, the idea of co-benefits, all of that has to be part of our governance structure. And at the moment, it doesn't have a home. It doesn't have a legal home or an institutional home.

To entrepreneurs, one and a half points. The one point is, I'll just repeat very briefly, which is, I think, look for these opportunities where you can make, in particular, adaptation and resilience combined with development opportunities and provide services to rural communities in particular. And the half point is, perhaps stay away from the offsets market.

Okay. So at the end of every episode, we generally try and do a good deed, something charitable in nature or help an entrepreneur who's trying to build in this space. So what we typically do is ask people to commit any amount of money.

It could be as small as, you know, like whatever. It doesn't matter. Each of you will pick a charity that you like.

And then we'll get the audience to vote. What it does is gives that charity a lot of visibility. And then we put out a video of the work they've done using the money we kind of like committed.

So I'm just going to go around and ask you guys, commit some money. Don't think about the quantum of money as the thing. 10 lakhs.

Okay. Same. You might have to do in-kind contribution.

Yeah. Works. So 10, 10.

I'll do 30. So that makes it 50. We'll get the audience to pick.

And you guys will come up with the charities that will be in the poll. But thank you guys so much. Thank you for taking all this time.

Thank you for doing this. I think this will help a lot of young people who have heard about climate change, but up until today did not understand it in the manner that they do after today. So thank you.

Thank you for hosting us. I hope that was okay. Would you take a photo?

Yeah, of course.